

Yes/No to How Much: Designing a Quantitative Dengue NS1 Test

De novo protein binder design against dengue NS1, and a calibrated benchmark for judging designs

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Abstract. Dengue rapid tests answer one question: is NS1 present or not. They cannot say how much, so they cannot say whether a patient is *improving* or *deteriorating*. We set out to design *a de novo protein switch* that binds dengue NS1 and reports its concentration as light, using RFdiffusion, ProteinMPNN and co-folding validation for the pipeline. Before ranking any design, we calibrated the acceptance gate against known answers: a protective anti-NS1 antibody with a solved crystal structure, and a non-cognate nanobody with no reason to bind. The gate ended up being inverted. Confidence was anti-correlated with measured epitope contact (Spearman rho = -0.888, n = 15), and the real antibody failed every seed while the decoy cleared the confidence thresholds in three of five. A published antibody predicted against its own cryo-EM epitope recovered none of its 16 residues in 10/10 runs at unchanged confidence. Ranking on epitope contact instead yields *two binder arms* at *non-overlapping NS1 sites*, one 84% conserved across DENV1-4, with measured geometry compatible with split-reporter complementation and a construction-ready specification. We release the scoring harness, the calibration justifying it, and a corrections ledger. Still, the switch awaits assembly: a predictor that cannot locate a published epitope cannot adjudicate a designed one. What this work delivers is a filter whose behaviour on this target we measured rather than assumed, and the calibration data that shows it.

1. Introduction

1.1 The problem

1.2 Importance, tractability, and neglectedness

1.3 What we set out to build, and what we asked

2. Literature review

2.1 Current dengue diagnostics

2.2 Quantitative NS1 detection already exists

2.3 Protein switches and design tools

2.4 The gap

3. Theory of change

4. Methodology

Definitions

4.1 Target, epitope, and what we chose not to design against

4.2 The generation pipeline

4.3 Controls, and what they broke

4.4 Conservation, specificity, and switch geometry

4.5 Reproducibility and what is deposited

5. Results

5.1 The metric ran backwards

5.2 Three more ways the pipeline misled us

5.3 What the project delivers

5.4 What we withdrew

6. The proposed test

6.1 The format

6.2 Why the format follows from the architecture

6.3 Quantification

6.4 What would kill it, and what it would cost

7. Discussion

7.1 Interpretation

7.2 Limitations

7.3 Future work

7.4 Confidence and what it would take to change our mind

8. Conclusion

Author Contributions

AI Disclosure

References

Appendix

A1 Design shortlist

A2 Gate calibration:

A3 Predictor mislocation:

A4 Steric re-feasibility against intact NS1

A5 Rescue campaign: all fifty predictions

[A6 Charge-variant series](#)
[A7 Second-arm conservation against DENV1-4](#)
[A8 Direction B burial analysis](#)
[A9 Switch geometry](#)
[A10 DENV1-4 alignment](#)
[A11 Provisional target product profile](#)
[Analytical performance](#)
[Operational performance](#)
[A12 Corrections ledger](#)
[A13 Reproduction instructions](#)
[A14 Cross-validator agreement](#)
[A15 Monomer versus dimer: a registered prediction](#)

1. Introduction

1.1 The problem

A dengue test in the field today returns a binary bit of information: yes, or no. While that is alright, our project centers on making it more quantitative.

Dengue infects around **390 million** people a year. About 96 million become ill enough to seek care [6], roughly half the world's population lives where transmission can occur, and in the first seven months of 2025, the World Health Organization logged over 4 million cases and more than 3,000 deaths [27]. Most patients recover, however a small fraction deteriorate sharply, usually days after the fever breaks, and this deterioration can lead to serious complications.

The virus does leave something to monitor, though. Infected cells shed a protein called **NS1** straight into the blood pretty early before the antibodies appear, and its concentration rises and falls over the illness. Now this is quite significant: NS1 tracks with viral load on the same day of illness and declines as the infection resolves [25], so measuring it more than once tells you *which direction* a patient is heading. A strip that reads positive on day two and positive on day five tells you nothing new, which is what makes the current test uninformative.

Measuring how much is not really hard, but it is not portable. A quantitative NS1 assay needs a plate reader, cold-chain reagents and a trained operator, which in practice means a district hospital laboratory. Hence, we need a test that reaches the patient on a paper strip, and a strip answers one question.

1.2 Importance, tractability, and neglectedness

Importance: Why does the test need a building? It is because of how the assay is built, and not because the measurement is delicate. An ELISA immobilizes an antibody on a plate, washes away everything that did not stick, and then adds a second labelled

antibody to report what stayed. All those steps exist to separate bound from unbound. Now, a molecule that produces light only when bound would never face that problem, and there would be nothing left that needs a laboratory.

However, one thing should be taken carefully. We began believing NS1 concentration predicts who becomes dangerously ill, but the literature does not support it. Some cohorts find high NS1 in shock, others in mild cases, and the disagreement tracks confounders the studies handle differently. So instead, we will try to just quantify it rather than give a specific direction, and this is still something a yes-or-no can never report.

Tractability: This was harder than the published record suggested. A binding site worth targeting, we realized, has to be conserved across *all four serotypes*, or the test fails on whichever one arrives. Plus, a designed binder has nothing dependable to grip onto. On NS1 those requirements point at the β -ladder: conserved, rigid, and flat. But flat is a problem, as de novo design performs best against globular targets whose pockets and grooves give a binder something to wrap around, and an extended β -sheet offers almost none of that. Our designs converged on a pose *less reliably* than the literature predicts, and the confidence scores used to sift them carried no usable signal here. That second problem stopped being an obstacle and later became the finding.

Neglectedness: Designed protein switches (single molecules that both recognise a target and announce it) have been demonstrated against SARS-CoV-2, HER2, BCL-2, cardiac troponin-I, parathyroid hormone and glucagon. The list is a pattern rather than a catalogue, as every one is an analyte already measurable in a well-equipped hospital, where a switch competes on convenience. Not one is a disease (especially a neglected tropical disease) whose defining problem is that the laboratory is absent altogether, which is where recognition and reporting in the same molecule is worth the most.

1.3 What we set out to build, and what we asked

Hence, this is what we set out to design. Two small proteins, each built from scratch to grip a different spot on NS1, each carrying half of a light-producing enzyme. When apart, nothing happens, but when both grip the same NS1 molecule at once, the halves are pulled together into a working enzyme and the sample glows, brighter the more antigen is present. This only required one tube, one drop of blood, no washes, and no plate reader, because the separation problem was never there.

Phase one, designing and characterising the two binders, is finished and is what this report covers. **Phase two**, assembling them into the switch, is fully specified and but deliberately not built, on the evidence that phase one produced.

Three questions organise this work, and Section 7 will answer them in order.

1. Can a de novo binder be designed against the NS1 β -ladder epitope (residues 261 to 305) conserved across all four serotypes and not cross-reactive with relatives like Zika?

2. Can two binders occupy non-overlapping sites on NS1 at once, with the geometry a split reporter needs?
3. Do the standard confidence metrics for designed complexes identify correct binders on this target?

2. Literature review

2.1 Current dengue diagnostics

RT-PCR remains the reference standard for acute-phase dengue detection, and its limitation is largely practical: it needs specialised equipment and trained operators, which restricts it to centralised laboratories.

NS1 antigen capture is the alternative, in two forms that differ in what they can tell you. The NS1 ELISA is quantitative and standardised, but heterogeneous: a capture antibody immobilised on a plate, sample incubation, wash steps, a separate labelled detection antibody, substrate, and a plate reader. Those steps are why it needs a laboratory. The rapid diagnostic test collapses that workflow into a strip and gives a binary answer in return.

The performance of that trade is well documented and unflattering. Across field studies, NS1 rapid tests report sensitivity from 61.0% to 87.4% against RT-PCR, with specificity consistently between 97% and 100% [23, 16, 25]. In a Bangladesh outbreak cohort, positive predictive value was 91% but negative predictive value only 83% [23]. The clinical implication is consistent: a positive strip is trustworthy, while a negative strip cannot rule out dengue.

The mechanism behind that asymmetry matters for what follows. Sensitivity is not uniformly lower; it falls in predictable circumstances: when DENV-reactive IgG is present, in secondary infection, and later than three days after symptom onset [25]. In secondary infection, falling NS1 levels alongside rising anti-NS1 IgG are consistent with immune-complex formation, although IgG-mediated clearance provides an alternative explanation for the same pattern [18]. Either would reduce the amount of free NS1 available to an antigen-capture assay. Dissociating immune complexes with acid does help, raising sensitivity from 63.2% to 72% in one Bangkok cohort in which 92.4% of dengue cases were secondary infections [15], but treatment also disrupts the native NS1 state [1].

That suggests a design hypothesis more specific than maximising affinity. If diagnostic antibodies and patient IgG compete for overlapping NS1 surfaces, a binder deliberately targeted to a different, structurally suitable surface could reduce that competition. This remains a prediction, not a demonstrated property of our binder. Testing it properly would require an epitope-level map of the dominant human anti-NS1 response, which we did not have when the target window was locked; our epitope was chosen on rigidity and DENV1-4 conservation (Section 4.1), and Section 7.3 sets out what selecting on immunodominance would require.

2.2 Quantitative NS1 detection already exists

We began this project believing that quantitative NS1 measurement was itself the unmet need. That was wrong, and correcting it narrowed the claim usefully.

A substantial biosensor literature already achieves quantitative NS1 detection at very low concentrations. A DNA aptamer electrochemical sensor reports a 22 pg/mL detection limit in undiluted human serum [5]. Capacitive aptasensors report detection limits in the tens of femtograms per millilitre across a linear range from 10 pg/mL to 1 µg/mL [3], and a gold-nanoassembly format about 1.28 pg/mL colorimetrically and 30 fg/mL electrochemically [13].

The gap is therefore not analytical sensitivity but deployability. These platforms require fabricated electrodes, synthesised nanoparticles, or electrochemical instrumentation, the class of equipment that makes translation outside a laboratory harder. A second finding is worth carrying forward: a low-cost dual dengue and Zika aptasensor proved unable to discriminate between the two analytes [4]. Flavivirus cross-reactivity is therefore a demonstrated failure mode in a real device, not only a theoretical concern, and it justifies checking serotype coverage and off-target behaviour rather than assuming specificity.

2.3 Protein switches and design tools

The architecture this project builds on comes from the LOCKR and lucCage lineage of designed protein switches, in which target recognition is coupled directly to a luminescent output [14, 22]. This removes the need for a separate detection step because binding itself changes reporter activity.

One precedent is close enough to be decisive. The lucCageRBD SARS-CoV-2 sensor incorporated LCB1, a computationally designed de novo miniprotein binder, into the lucCage scaffold and reached a 15 pM limit of detection with a luminescent signal about 50-fold above background [22, 8]. That is the route this project takes: design a de novo binder against a viral antigen, couple recognition to a split reporter, and obtain a luminescent readout. It substantially de-risks the architecture and means the novelty here is the target and design pipeline rather than the switch itself. Bioluminescent sensing has also been read directly in blood plasma using a smartphone, providing a separate precedent for the proposed camera-based readout [2].

The design pipeline still carries a limitation worth stating before the results. RFdiffusion has demonstrated de novo binders against several conventional protein targets [26], but that does not establish that the same workflow will perform equally well against every surface geometry. The NS1 β -ladder presents a relatively flat, extended β -sheet face. Whether the generator and the co-folding metrics used to rank its outputs transfer cleanly to that target is therefore an empirical question, and Section 5 reports what happened when we tested it.

2.4 The gap

Nothing in the preceding paragraphs is invented here. Quantitative NS1 detection, protein-switch architectures, bioluminescent point-of-care readout, and protein sensors for infectious-disease targets all already exist. A bioluminescent sensor has even been applied to dengue: LUMABS was demonstrated on antibodies against dengue virus type 1 [2]. That is serology, however, rather than direct detection of circulating NS1 antigen.

The claim is correspondingly narrow. In our literature search, we found no published de novo protein switch against NS1 antigen, and no published application of an RFdiffusion, ProteinMPNN and AlphaFold-class binder-design pipeline to this target; LCB1 predates RFdiffusion [8, 26]. We also found no published example of such a campaign being structured as a reproducible, target-agnostic benchmark. The project therefore does not claim a new sensing principle. It tests whether an existing protein-switch strategy can be extended to a clinically relevant dengue antigen using a newer generation of computational design tools, and records where that pipeline succeeds and fails.

3. Theory of change

The pathway from this work to a quantitative dengue test runs through wet-lab validation of designed molecules, and it is worth being precise about what that requires and when.

The chain runs from designed binders to a deployed test in four stages, and the project sits at the boundary between the second and the third. The impact, if the later stages hold, is serial NS1 measurement at the point of care, where today there is a single yes-or-no strip, with no potentiostat, no washes and no cold chain.

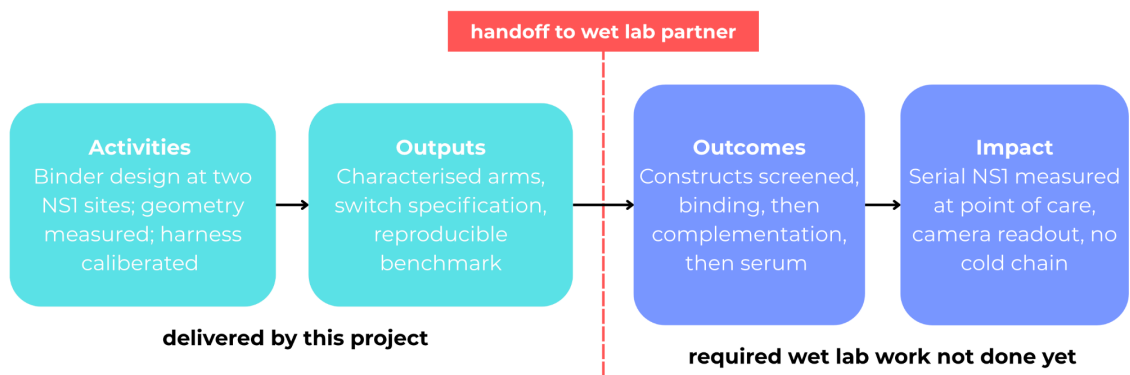


Figure 1. the impact pathway, with the handoff point marked. Teal boxes are delivered; purple boxes require wet-lab work.

What a partner receives today is a scoring harness, the crystallographic calibration that justifies its selection criterion, and two characterised binder arms. That package exists now and does not depend on phase two. The immediate ask is small: binding assays on the two free arms against recombinant NS1, the cheapest experiment that could falsify

this work. Phase two assembles the constructs the specification defines: arm 1 is the β -ladder binder fused through an 8 to 10-residue linker to SmBiT, arm 2 the wing and domain-I binder fused to LgBiT, giving 8 to 24 constructs across the open choices of pairing, linker and orientation. Assembly is string concatenation from measured geometry, so the concatenation step adds no new predictor dependence, though the geometry it concatenates carries the one described in Section 5. Complementation is then measured alongside controls: free arms, a non-complementing same-epitope pair, and reporter-only background.

The groups positioned to act are identifiable. Diamond's laboratory at Washington University works at the flavivirus and protein-switch intersection; the Institute for Protein Design built the tools this pipeline runs on; OUCRU, ICDDR,B and ICMR-NIV Pune are natural evaluation partners and sources of the NS1 performance data cited here; and the LSHTM International Diagnostics Centre, which developed the dengue pre-vaccination screening profile, is the right group to ask what a quantitative one requires. A local connection in Amritsar is furthest along.

Five preconditions stand between this work and a deployed test, and we state them because a pathway that hides its gates is not a pathway. Only one of them has any evidence behind it today.

Table o. *The five preconditions between this work and a deployed test, with the evidence for each today and the experiment that would resolve it.*

#	Precondition	Evidence today	Resolved by
1	The designed binders bind their intended sites	None	Binding assay on free arms
2	The two arms bind simultaneously, not competitively	Computational: geometry measured on superposed structures (Section 5.3)	Complementation screen
3	Proximity produces a signal with usable dynamic range	None	Dose-response on assembled constructs
4	The format survives serum, including immune-complex masking	None	Spiked-serum panel
5	A manufacturing and regulatory route exists for a novel protein reagent	None	Outside this project's scope

We claim nothing beyond row 2, and that row is computational rather than experimental. Two further caveats belong here rather than in the limitations: no partner is yet secured, and the criterion used to select these candidates was defined after seeing the calibration data rather than before it.

The transferable contribution does not depend on any of this. The calibrated harness is not dengue-specific: any campaign against a target with a solved reference structure can run the same control, and the finding that a confidence metric can be anti-correlated

with correctness is a claim about method rather than about NS1. If the binders fail, that survives.

Two things this project is not. It is not a severity-prediction tool, because the NS1-severity literature is contradictory and we make no prognostic claim. It is not the first quantitative measurement of NS1, because sensitive sensors already exist. The claim is narrower: a homogeneous instrument-free architecture, a de novo design route to it, and an openly released benchmark for judging designs along the way.

4. Methodology

Definitions

Table 1.

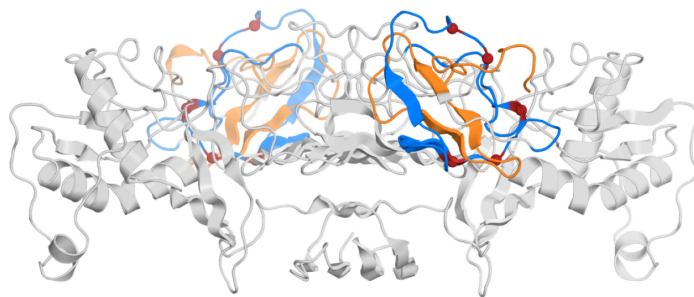
Term	Meaning
ipTM	Predicted confidence that two proteins are correctly docked together; higher is not always more correct (Section 5.1).
Interface pAE	Predicted positional error at the binding interface; lower values indicate higher predicted confidence.
pLDDT	Per-residue confidence that the local structure is modelled correctly.
BSA / burial	Surface area hidden from solvent when two molecules bind, used here as a proxy for contact size.

4.1 Target, epitope, and what we chose not to design against

The target throughout is: DENV2 NS1, PDB 9W22 chain A, residues 1-346, which is the ordered monomer [29]. All design and validation uses the monomer, and the dimer appears only for epitope-accessibility questions (Refer to Section 4.3).

The project began with six candidate biomarkers for neglected tropical diseases, but we narrowed to NS1 alone. We excluded the other five due to structural reasons. For example, LAM is a glycan and offers no protein interface, rK39 and rTcZ are antibody-capture antigens rather than circulating analytes, and PfHRP-2 is intrinsically disordered. Designing a rigid binder requires a rigid, exposed, circulating protein surface, and NS1 was the only clean candidate that provides one.

a Secreted NS1 dimer: both target faces sit on the same surface



b The two regions are interdigitated strands of one β -sheet

β -ladder epitope
residues 261-305

tip region 311-352
avoided: host mimicry

● hotspot residues
269, 272, 274,
294, 296

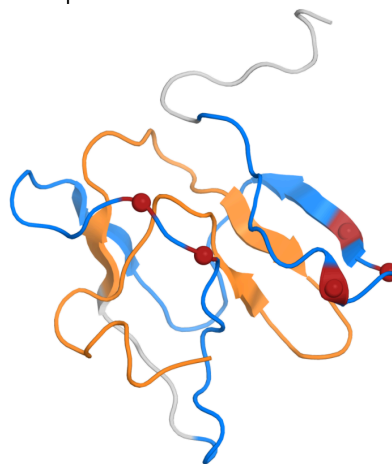


Figure 2. the design target. (a) DENV2 NS1 dimer (PDB 9W22): the β -ladder epitope, 261 to 305, in blue and the avoided tip region, 311 to 352, in orange. Both faces lie on the same surface, which is what makes a two-arm sandwich possible. (b) The same regions on one protomer: interdigitated strands of a single β -sheet, so burying one buries the other. Red spheres mark the five nominated hotspots.

Within NS1 we chose the **β -ladder patch**, residues 261-305, with nominated hotspots 269, 272, 274 (cluster A) and 294, 296 (cluster B). The patch is rigid, surface-exposed, and conserved across DENV1-4. We deliberately avoided the tip region, 311-352, which carries host-mimicry motifs implicated in cross-reactive pathology. For this reason, a binder engaging it risks both false positives and an unattractive safety profile.

A second, **wing / domain-I site** (~104-125) was carried as a stretch objective. It later became the second arm of the switch (Section 5.3).

The design plan was two-phase from the outset:

- generate and characterise binders at two non-overlapping NS1 sites (phase one),
- fuse them to split-reporter fragments and model the assembled switch (phase two).

Everything in Section 4.2-4.3 is phase one, while Section 4.4 delivers the phase-two specification. Building the constructs it defines is properly under phase two.

This choice is where the project's central constraint originates, as the epitope and the avoid-window are interdigitated strands of a single β -sheet, 3.0-4.3 Å apart. That is what Section 5.4's refuted safety criterion is about, and it was known from the start. We simply did not realise how binding a constraint it would turn out to be.

4.2 The generation pipeline

There are three stages to any generation, each in its own version-pinned conda environment on a single NVIDIA A40 (CUDA 12.4). The environments are kept separate because their dependency stacks conflict [26, 9].

Table 2. *The three-stage generation pipeline: tool and role at each stage.*

Stage	Tool	Role
Backbone generation	RFdiffusion	binder backbones against the epitope hotspot set
Sequence design	ProteinMPNN	sequences for each backbone
Validation (co-folding)	Boltz-2 2.2.1	binder-NS1 complex prediction and scoring

Every metric comes from Boltz-2 2.2.1, under *one harness* used identically for designs and controls: `recycling_steps 3, diffusion_samples 1, --use_msa_server, --no_kernels, seeds 1-5` [21]. Reproducing any number requires this exact harness, and ipTM is not comparable across different diffusion-sample or recycling settings. ColabFold/AF2-multimer was used only for the cross-validator check which is in Section 5.1 [12].

Now, three conventions govern every number that is reported below.

Contact definition: Refer to Appendix A13.

Design nomenclature: Designs are named by target site and rank on the primary acceptance criterion, such as B1, B2, etc. at the β -ladder site (261-305) and W1, W2, etc. at the wing / domain-I site. These are numbered by epitope-fidelity rank within each. This is not an indicator of confidence, as B1 ranks first on epitope fidelity and ninth on ipTM.

Five-seed protocol: Every construct is co-folded under five independent seeds and scored per seed, so all per-construct quantities are distributions rather than single values: the epitope-window rate (seeds of 5 placing ≥ 4 contacts in 261-305, which is not the primary criterion of Section 4.3), the per-seed hotspot frequency, and the consensus footprint. Hotspot engagement is always a per-seed frequency and never a union across seeds, which would count a hotspot as engaged if any single run touched it and overstate reproducibility. Figures and deposited columns labelled on-epitope report this window rate.

Burial convention: Buried surface area is reported target-side, binder-side, or per side, and these must be taken separately. Every burial figure here is target-side, filtered to residues 261-305. In fact, conflating the conventions once made an epitope subtotal appear to exceed its own total. Cropped-target constructs are recorded as untestable against a criterion referencing residues outside the crop, not as passing.

4.3 Controls, and what they broke

This is essentially the substance of the methodology. Four controls were run that a standard binder-design protocol does not require, and three of them *invalidated* a component of the pipeline.

The first control calibrates the acceptance gate against other known answers. We locked that gate before generating anything, adopting it from globular binder design:

- pLDDT > 0.8
- ipTM > 0.7
- interface pAE < 5 Å
- at least three contacts inside 261-305. One clarification matters for reading the calibration table, which is that the gate_pass column there records only the three confidence components and keeps contact in a separate column, so a construct can appear as passing while making zero epitope contacts and both facts remain visible on the same row.

Rather than applying it, we first tested whether it could sort constructs whose status we already knew: **2B7**, a protective anti-NS1 antibody with a deposited crystal structure; **cAb-Lys3**, an anti-lysozyme VHH with no reason to bind NS1; and one of our own designs [7, 17, 10]. The same harness was used for all three constructs across all 15 predictions.

The gate ended up inverting (Section 5.1). Its ipTM > 0.7 and pAE < 5 Å thresholds are reported as invalid absolute cutoffs for this target. pLDDT > 0.8 is retained only as a fold-sanity guard, and since all three constructs clear it, it does not discriminate either.

What separates the negative control from both real binders is *where* the binder lands, not really how confident the model is:

- **Primary criterion (arm 1, β-ladder):** ≥4 heavy-atom contacts within 261-305 and ≥2 of the five nominated hotspots with ≥1 from cluster A, assessed per seed and reported as a pass rate over five seeds. This is stricter than the epitope-window rate of Section 4.2, so the two are reported separately throughout. Only 2 of the 50 rescue predictions meet it, B1 at seed 1 and B4 at seed 4.
- **Primary criterion (arm 2, wing / domain-I):** the criterion above is defined over the β-ladder, so it says nothing useful about a binder that targets somewhere else. Arm-2 candidates are selected on consensus reproducibility instead: how large a contact core recurs across all five seeds, and how well that core is conserved across DENV1-4. Like the arm-1 criterion, it was defined after seeing the data.
- **Secondary (relative confidence):** ipTM and interface pAE reported relative to the 2B7 reference on the same predictor (five-seed means: ipTM 0.460, interface pAE 6.43 Å), never against absolute cutoffs.
- **Fold guard:** pLDDT > 0.8.

This criterion was defined after seeing the calibration data. It describes the observed binders rather than independently testing them, and a held-out validation would require new designs scored blind. We state it wherever the criterion is applied, because a post-hoc criterion presented as pre-registered would be the single most misleading thing in this report.

The second control tests steric feasibility against the intact target. Designs generated against a cropped epitope fragment can score well on every confidence metric and still be impossible on the whole protein. Each cropped-target complex is superposed back onto full-length NS1 via its own target fragment (C α RMSD 0.95-1.00 Å) and tested against the rest of NS1 for heavy-atom pairs below 2.0 Å. No confidence metric flags this. Any epitope-focused protocol that crops its target must run this test as a mandatory filter.

The third control asks whether the predictor can find an answer it has already been given. We predicted a published antibody (3G2) against NS1 and asked whether the predicted interface recovered its known cryo-EM epitope. Then, two alternative explanations for failure were specified in advance and tested by measurement: whether the epitope residues were structurally absent or disordered, and whether they were occluded in the physiological dimer, and *neither* survives.

The fourth control tests cross-validator agreement. Five designs were re-scored under AF2-multimer and Boltz-2 with identical inputs, to test whether the calibration result was one model's idiosyncrasy (Appendix A14).

Two of these controls destroyed work already done: the first invalidated the ranking criterion, the second an entire design campaign.

4.4 Conservation, specificity, and switch geometry

DENV1-4 conservation and DENV/ZIKV specificity are computed at 4.5 Å-defined interface positions against alignments whose numbering was validated before use. Whole-NS1 questions use a four-serotype alignment (ns1_aln.fasta, 352 ungapped columns), validated against 9W22 chain A at offset 0 with 335/346 = 96.8% identity, and all 45 positions in 261-305 reproduce the trusted β -ladder table exactly.

One lookup trap cost us a result. Two narrower alignment tables span only 261-305, and querying them for a residue outside that window returns "not conserved" silently, producing a spurious 0/19 conservation result for the second-arm site whose correct value is 16/19. Any site outside 261-305 must be scored against the whole-NS1 alignment. The failure mode is invisible: a wrong answer, not an error.

ZIKV residues are strain-dependent: 2 of 45 β -ladder positions disagree between the stored specificity table and the NCBI isolate used (AMC13911.1). All ZIKV claims inherit this isolated dependence.

The architecture is a two-arm proximity sandwich, not a conformational switch. This is because two binders at non-overlapping NS1 sites are each fused to one half of a split NanoBiT luciferase (SmBiT on arm 1, LgBiT on arm 2), so signal arises from

antigen-driven co-localisation [22]. LOCKR/lucCage-style switches were considered in the start and then rejected- they require a designed closed state whose stability we cannot reliably predict, and Section 5.1 gives reason not to trust an uncalibrated confidence metric.

Geometric compatibility is assessed by superposing each arm's predicted complex onto a common NS1 frame and measuring, per seed-pair, the minimum heavy-atom separation between arms (2.0 Å threshold) and the C-terminus-to-C-terminus distance that sets the linker requirement. These are measurements on predicted complexes, not predictions of switch behaviour, and no signal is claimed from them.

Phase two is scoped rather than open. The specification fixes the construct space of binder pairs $\times$ linker lengths $\times$ reporter orientations, and the control set is enumerated: free unfused arms, a non-complementing same-epitope pair, reporter-only background, and a linker series. What has not been done is assembling and modelling them. No switch result appears in Section 5 because none has been run, a scope boundary rather than a negative finding.

4.5 Reproducibility and what is deposited

Every finding in Section 5 is reproducible from the deposited artifacts and fixed harness detailed in Appendix A13, including the corrections ledger in A12.

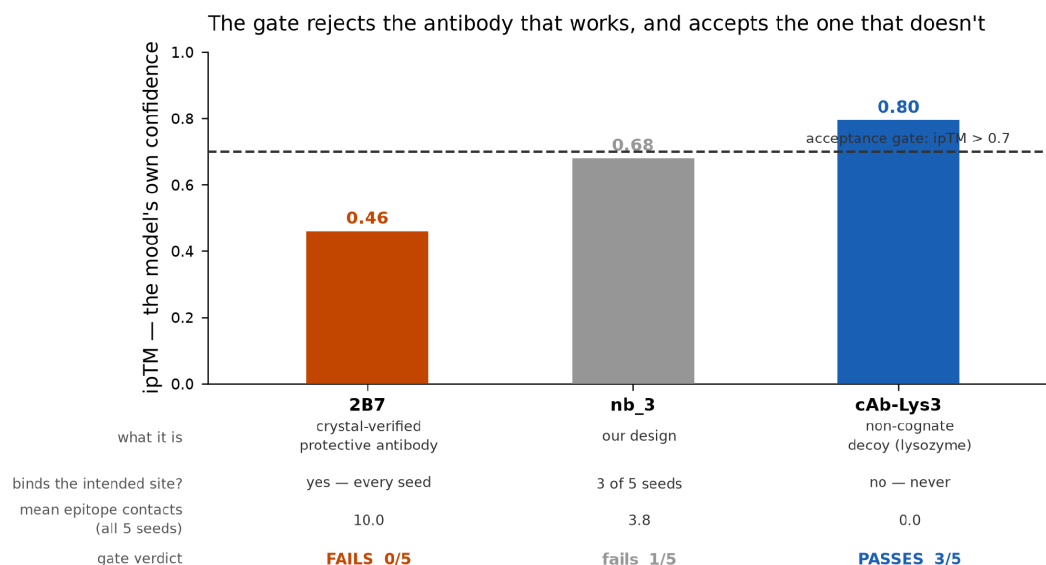
5. Results

5.1 The metric ran *backwards*

Calibrating the gate defined in Section 4.3 against **2B7**, **cAb-Lys3**, and **nb_3** (five seeds each) inverted the expected ordering

Table 3. Gate calibration against three constructs of known status, five seeds each. The antibody with a solved structure passes no seed; the decoy passes three. Gate pass reflects the three confidence thresholds only; epitope contact is reported separately in the adjacent column.

Construct	Status	Mean ipTM	Gate passes	Mean epitope contacts
2B7	Known binder (crystal reference)	0.460 $\pm$ 0.012	0 / 5	10.0 (10 every seed, within 261-305)
nb_3	Our design	0.680 $\pm$ 0.105	1 / 5	3.8 (7, 6, 0, 0, 6)
cAb-Lys3	Non-cognate control	0.795 $\pm$ 0.079	3 / 5	0.0 (0 every seed)



Means over all 5 Boltz-2 seeds against full-length DENV2 NS1 (9W22 chain A); nb_3's per-seed contacts are 7, 6, 0, 0, 6. Epitope contacts measured from predicted coordinates — a quantity the model never sees. Ten predictions total.

Figure 3. the three constructs against the $ipTM > 0.7$ gate line. The antibody with a solved structure scores lowest; the decoy with no reason to bind scores highest.

The antibody placed all ten of its epitope-window contacts inside residues 261-305 in every seed, recovering 18 to 20 of its 20 known contact residues overall, and failed the gate every time. On the other hand, the non-cognate nanobody made zero epitope contacts in every seed and cleared the confidence thresholds three times. Across all fifteen predictions, ipTM was anti-correlated with measured epitope contacts:

- Spearman $\rho = -0.888$
- $p = 9.7 \times 10^{-6}$
- $n = 15$.

Gate-passing predictions averaged **0.0** epitope contacts, while gate-failing predictions averaged **6.27**.

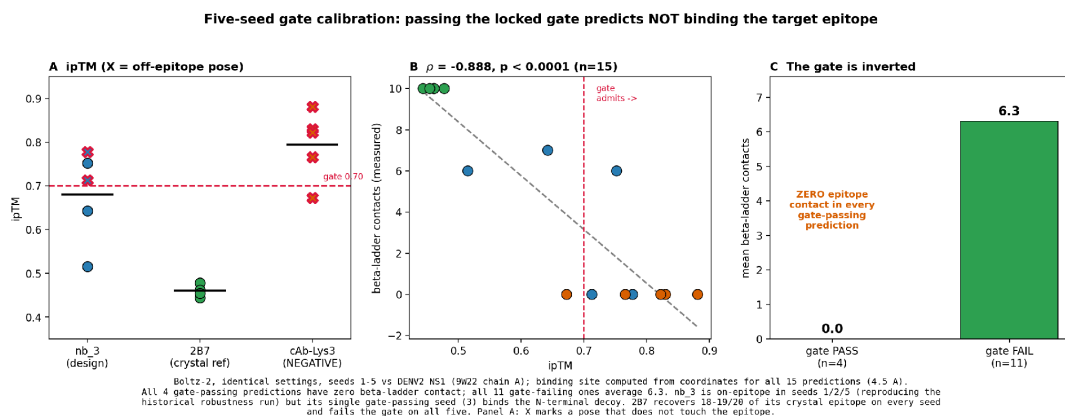


Figure 4. per-seed ipTM against measured epitope contacts across all fifteen predictions.

This is not only seen in one artifact; rather, when five earlier designs are re-scored under AF2-multimer and Boltz-2 with identical inputs, none clears the ipTM gate under either, and the two disagree sharply on the same structures, AF2 returning 0.14 to 0.20 where Boltz-2 returns 0.23 to 0.50. These five are three β -ladder backbones, our nb_3 construct and a PD-1 control, not the ten designs ranked below (Appendix A14).

All ranking below uses measured epitope contact instead.

We ranked the ten rescue-campaign designs both ways and compared the top three: the two shortlists share no molecules. The ipTM top three are on-epitope in 2 of 15 seed-runs, averaging 0.87 epitope contacts; the fidelity top three are on-epitope in 11, averaging 4.53.

- $\rho = -0.826$
- $p = 0.0032$
- $n = 10$.

Table 4. The same ten designs ranked by predicted confidence and by measured epitope fidelity.

Ranked by	Top three designs	On-epitope (of 15)	Mean epitope contacts	Mean overlap 2B7
Mean ipTM	W1, W3, B7	2	0.87	1.2
Epitope fidelity	B1, B2, B3	11	4.53	6.93

Design labels: B = β -ladder site, W = wing / domain-I site, numbered by epitope-fidelity rank within each site. Full mapping to pipeline identifiers in the supplementary table.

Table 5. Design labels used throughout, with target site and on-epitope frequency.

label	pipeline_id	target site	on_epitope_seeds_of_5
B1	onepi_pd_5_dl设计_1	β -ladder	5
B2	onepi_pd_16_dl设计_0	β -ladder	4

B3	onepi_pd_5_dlldesign_0	β -ladder	2
B4	onepi_pd_15_dlldesign_0	β -ladder	1
B5	onepi_pd_20_dlldesign_1	β -ladder	1
B6	onepi_pd_7_dlldesign_1	β -ladder	1
W1	seed5_pd_11_dlldesign_0	wing / domain-I	1
W2	seed5_pd_18_dlldesign_1	wing / domain-I	1
B7	onepi_pd_30_dlldesign_1	β -ladder	1
W3	seed5_pd_15_dlldesign_1	wing / domain-I	0

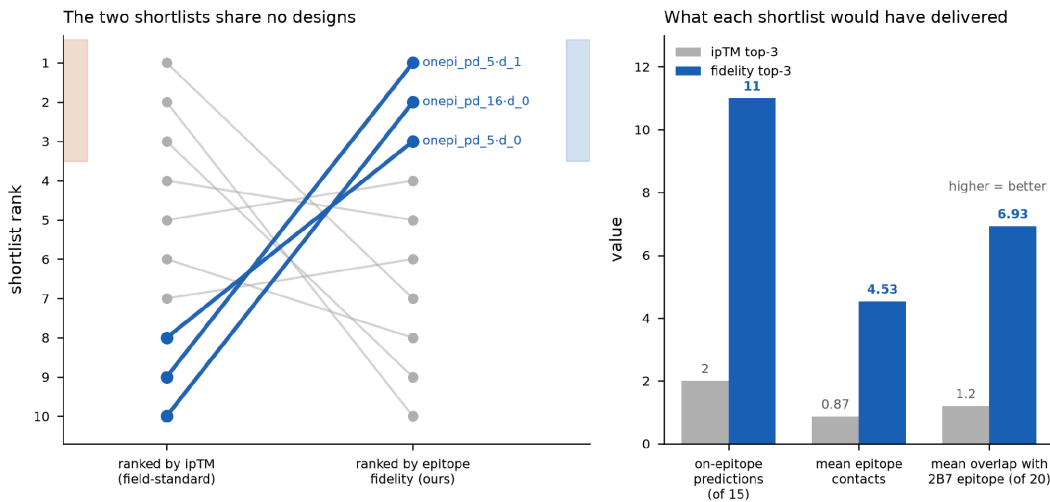
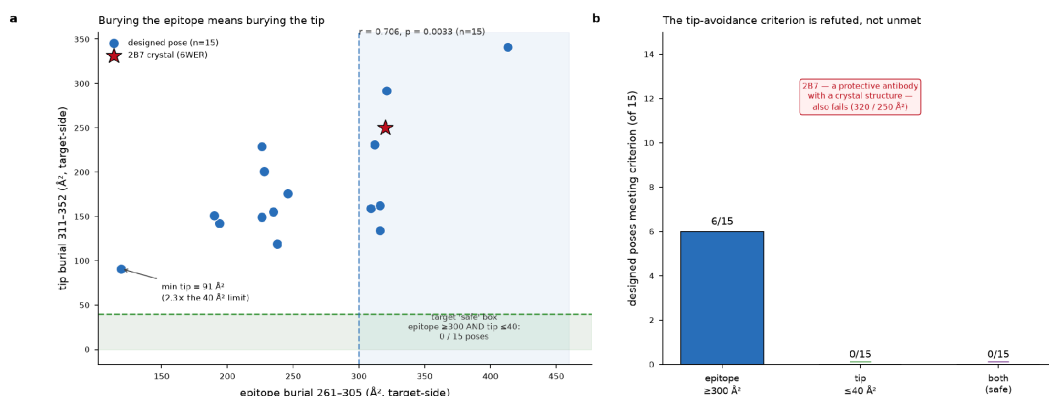


Figure 5. the same ten designs under both rankings, with the rank shift for each.

5.2 Three more ways the pipeline misled us

The designs generated against a cropped target do not fit the intact protein. The first campaign used a cropped NS1 target, the β -ladder alone, to keep the diffusion problem tractable. Tested against full-length NS1 (9W22 chain A, 1-346), zero of 32 were feasible, with 47 to 874 atom pairs closer than 2.0 Å against the parts the crop had removed. None of the design-time metrics gave any hint of this, and buried surface area is the clearest illustration: the design ranked first buried 1,703 Å² per side where the 2B7 crystal structure buries 602 Å², so the pipeline was rewarding an interface almost three times the size of a real one on a pose that cannot physically exist. Every campaign after this one was validated against the intact protein, though the sequences themselves were not thrown away, because the rescue campaign in Section 5.3 re-folded them freely against full-length NS1 and let the predictor find a new pose for each rather than holding the old cropped one fixed, which is why the same sequence can fail the superposition test in Table A4 and still be scored in Table A5.

Fig 15. Direction B re-derived on full-length NS1: the epitope and the avoid-window are one β -sheet



15 on epitope poses from 7 independently generated designs (Boltz 2, full-length NS1). All burial figures target-side. B, filtered to 261-305 (epitope) / 311-352 (tip). The two surfaces are interdigitated strands (3.0-4.3 Å apart), so epitope and tip burial are positively correlated. No pose comes within 2x of the 40 Å² tip limit; the protective antibody 2B7 fails it too — a criterion that excludes the one antibody known to work on this epitope is unsatisfiable by construction, retracts the earlier ~370 Å² burial ceiling and the 'five hotspots unsatisfiable' claim.

Figure 6. per-design clash counts against intact NS1.

A second failure concerns where the predictor puts things. We predicted 3G2, an anti-NS1 antibody with a cryo-EM epitope (PDB 8JKF) [20], against NS1 across ten runs, five binary and five ternary alongside our own binder. It recovered 0 of the 16 residues of its own epitope in all ten, and confidence never registered the error: ipTM 0.37-0.63, pLDDT 0.88-0.92. Two explanations would excuse this and both fail on measurement: only 1 of the 16 residues lies within 5 Å of a disordered or absent region, so the epitope was available, and it is not buried in the dimer interface, so it was not occluded. Where the predictor put 3G2 instead is a problem for us, because it made 5 to 6 β -ladder contacts in every run and the β -ladder is the surface our own designs target, so if the model leans toward that region by default it would inflate window rates for any binder and partly confound the metric we rank on. Two things limit the concern: the non-cognate decoy is not drawn there at all and lands on the N-terminus in all five seeds, so the pull is not universal, and our ranking depends on contact counts and consensus residues rather than the bare fact of landing inside the window. It does mean that a difference in window rate between two designs is weaker evidence than the consensus within a single design across its seeds.

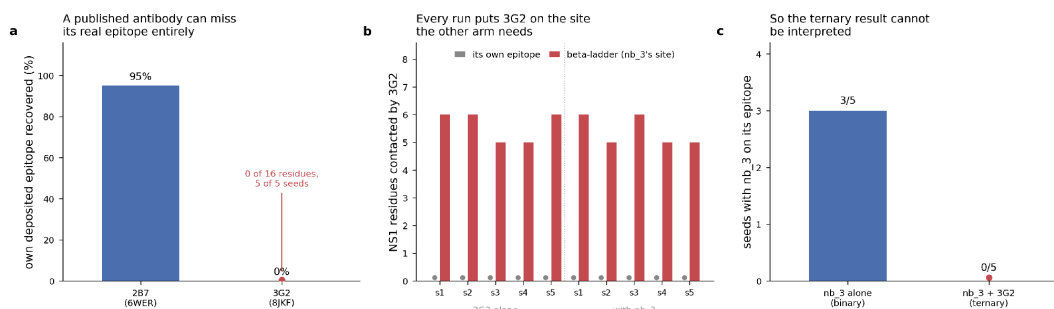


Figure 7. 3G2's predicted contacts against its known cryo-EM epitope, per run.

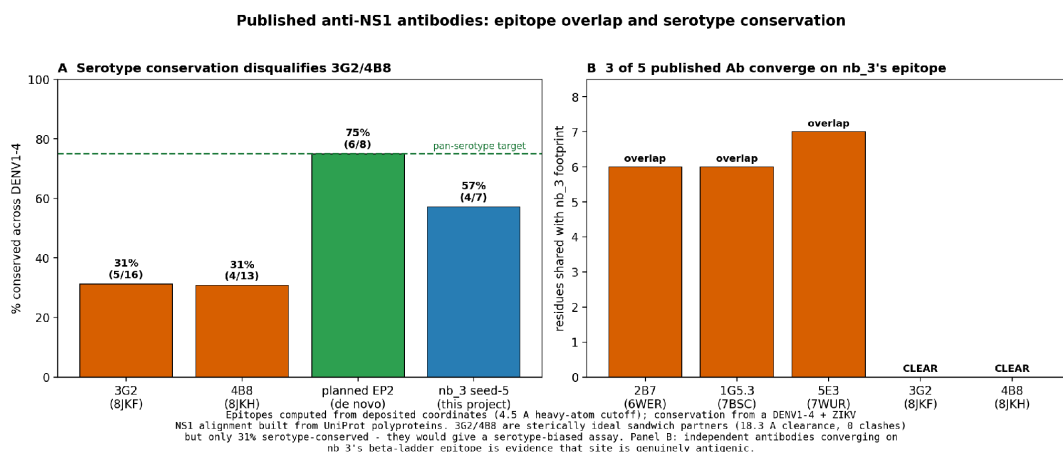


Figure 8. *predicted versus known epitopes for the published antibodies tested.*

The third failure is one we registered in advance. Before running it, we predicted that presenting NS1 as a dimer rather than a monomer would improve on-epitope frequency, the dimer being the physiological form. The opposite resulted: monomer gave 3 of 5 seeds on-epitope, the dimer 1 of 5. We report it because it was registered exactly as predicted; all ten predictions are deposited in Appendix A15.

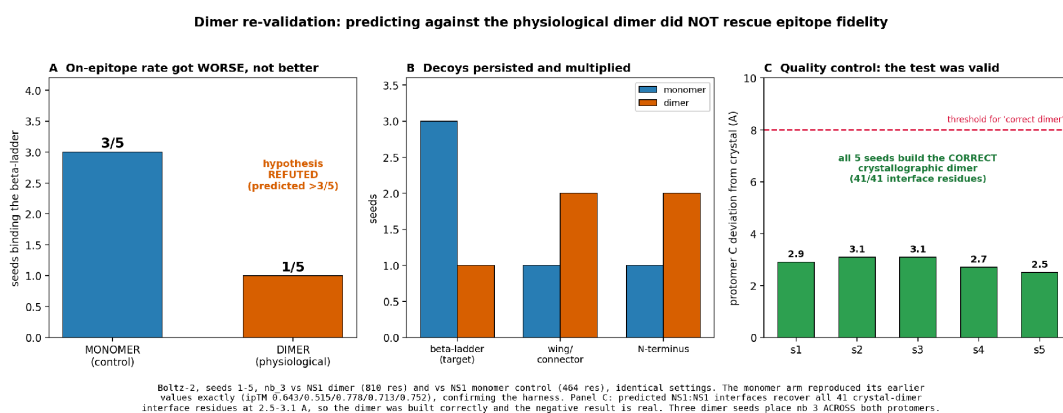


Figure 9. *on-epitope frequency under monomer versus dimer presentation.*

5.3 What the project delivers

Ranked by epitope-window frequency and mean epitope contact instead of ipTM, two candidates emerge from 50 predictions against the full-length NS1.

Arm 1, B1, targets the β -ladder. It reaches the epitope window in 5 of 5 seeds, with a mean of 6.2 epitope contacts (9.6 of 20 shared with the 2B7 reference) and residues 299, 303 and 327 contacts in every seed. The stricter primary criterion additionally demands two hotspots, and on that measure it passes in only 1 of 5, which is the gap between reaching a region and engaging the specific residues we nominated within it. The next design, B2, reaches the window in 4 of 5 seeds with 4.8 mean contacts and passes the primary criterion in none of them, but across those four on-epitope seeds it settles on 299, 303 and 327, which are the same three residues B1 converges on. Two designs

generated independently from different backbones landing on the same three positions is the strongest evidence we have that the site is real rather than an artefact of one trajectory.

Arm 2, W3, targets the wing / domain-I face. It was selected on consensus reproducibility at its own site, the arm-2 criterion of Section 4.3. A 19-residue contact core recurs in all five seeds, and 16 of those 19 positions are conserved across DENV1-4, an 84% realised footprint, superseding the 75% (6 of 8) quoted at design time for the planned window. Four positions in the footprint differ between dengue and Zika, so differentiation against the nearest cross-reactive flavivirus is available but not automatic.

Fig 14. Shortlist rescue on full-length NS1 — sequences recover where poses could not

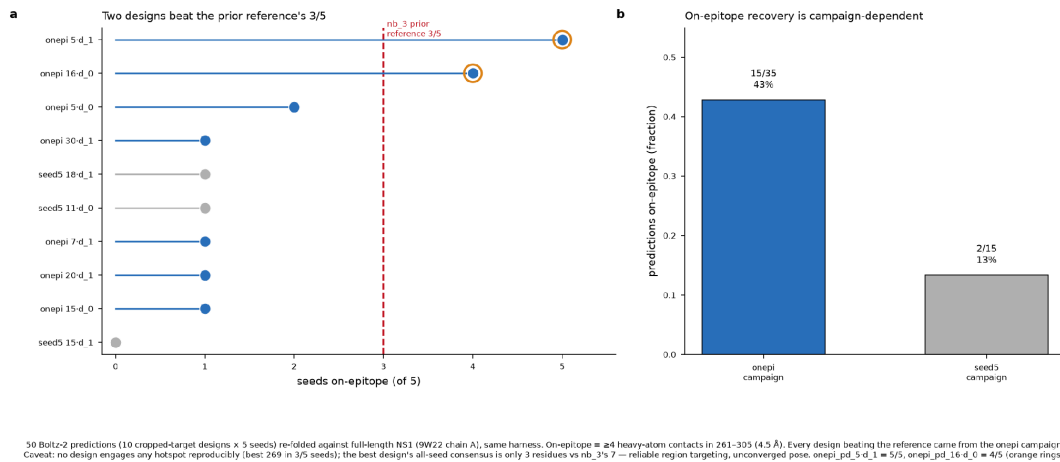


Figure 10. per-design on-epitope frequency and epitope contacts across the rescue campaign.

Fig 16. Second-arm lead: seed5_pd_15_d1design_1 binds the wing/domain-I site reproducibly

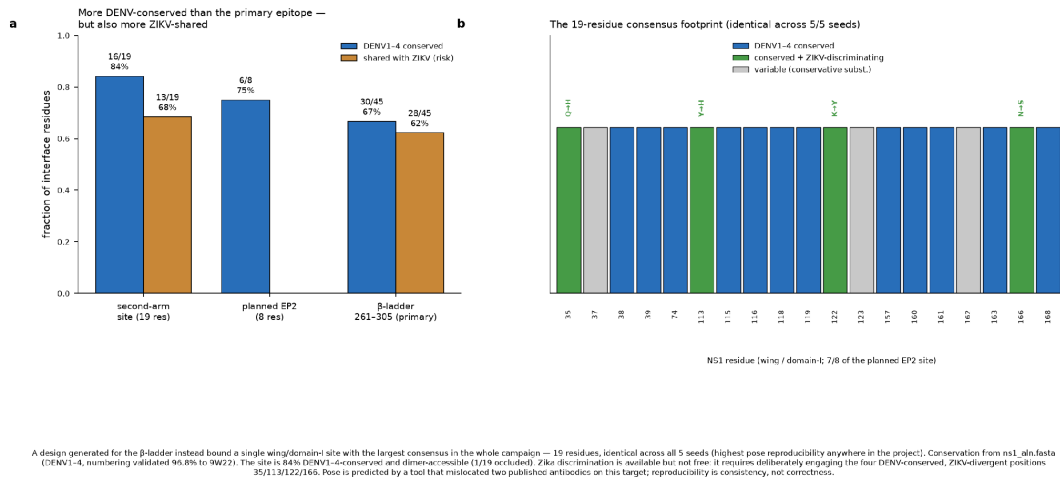


Figure 11. second-arm contact consensus mapped onto the DENV1-4 alignment.

Neither of these candidates were tested experimentally, they're only characterised.

Whether the two arms can occupy NS1 at the same time is what switch construction depends on. Across 25 seed-pairs, 22 clear a 2.0 Å steric threshold, and all 10 well-converged pairs clear it. A pair counts as well-converged when the two predictions' copies of NS1 superpose within 2.2 Å C α RMSD, since each folds the target independently and a pair disagreeing more than that cannot support a measurement between them. The median C-terminus separation is 30.6 Å across nine converged non-outlier pairs, setting the linker requirement at 8-10 residues per arm for NanoBiT. One pair, arm 1 seed 3 with arm 2 seed 5, comes within 2.42 Å: it clears the threshold with no severe clashes, but by the smallest margin. These are geometric measurements on predicted complexes, not a prediction of signal.

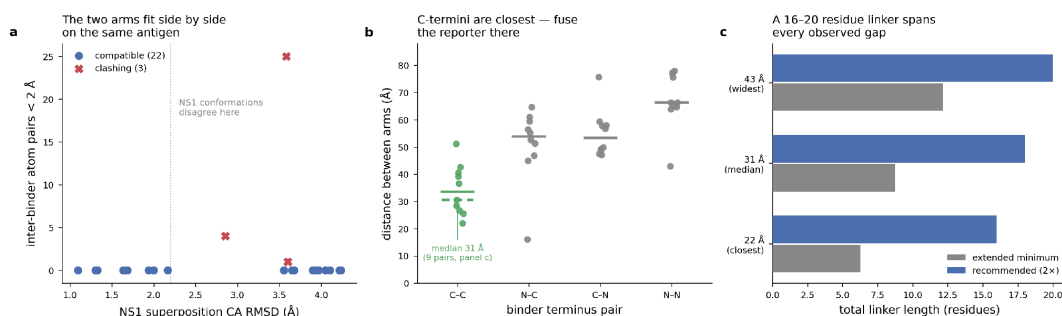


Figure 12. terminus separations and steric compatibility across seed-pairs.

Together these fix the construct space: two to three arm-1 candidates, B1 and B2 with B3 behind them if its 2 of 5 window rate is judged sufficient, and one to two arm-2 candidates, W3 then W1, together define 8-24 switch constructs ($2 \times 1 \times 2 \times 2$ at the low end, $3 \times 2 \times 2 \times 2$ at the high end). The two linker options are the 8 and 10-residue ends of the measured range, and the two orientations are which arm carries SmBiT. This specification is a deliverable of the present work; assembling and modelling the constructs it defines is phase two.

5.4 What we withdrew

Four things we set out to use did not survive the contact with measurement.

The first withdrawal is a safety criterion we set before designing anything. We required that a binder bury at least 300 Å² of the target epitope and no more than 40 Å² of the tip region, which carries host-mimicry and false-positive risk. Across 15 independent on-epitope poses, none met both, and the minimum tip burial was 91 Å², more than twice the limit.

The two burials are positively correlated ($r = 0.706$, $p = 0.0033$, $n = 15$), as the epitope and tip are interdigitated strands of one β -sheet, so burying more of one buries more of the other. What settles it is the crystallographic reference: 2B7 buries 320 Å² of epitope and 250 Å² of tip, failing our criterion while being a real, protective antibody. The criterion is refuted and withdrawn. Two further claims from the same analysis were retracted on measurement (detail in the Appendix).

Table 6. *The pre-registered burial criterion tested against fifteen on-epitope poses and against 2B7.*

Poses meeting epitope burial $\geq 300 \text{ \AA}^2$	6 / 15
Poses meeting tip burial $\leq 40 \text{ \AA}^2$	0 / 15
Poses meeting both	0 / 15
Minimum tip burial observed	91 $\text{\AA}^2$ (more than twice the limit)
Correlation, epitope vs tip burial	$r = 0.706$, $p = 0.0033$, $n = 15$
2B7 crystallographic reference	320 $\text{\AA}^2$ epitope / 250 $\text{\AA}^2$ tip - fails the criterion

The second withdrawal is a mechanism we had proposed for ourselves. We expected that raising CDR net charge would drive epitope engagement, and had called charge the one structural lever still available to us. Testing it across four variants spanning four charge states, five seeds each for 20 predictions, found nothing: Spearman $\rho = -0.137$, $p = 0.564$, $n = 20$. The best poses at +1 and +3 are indistinguishable, both reaching 8 contacts, while +5 never exceeds 4. The lever does not exist, and the recommendation built on it is withdrawn.

The third is a shortcut we wanted to take. A published antibody as the second arm would have removed the hardest remaining design problem, and 3G2 (PDB 8JKF) fits geometrically: 18.3 $\text{\AA}$ clearance, no clashes. What disqualifies it is conservation. Only 5 of its 16 epitope residues (31%) are conserved across DENV1-4, and the alternative, 4B8, is no better at 4 of 13. Either would read some serotypes better than others, defeating a quantitative test. Both are retained as methods-development models only.

The last withdrawal is a method rather than a claim. To check the tip region for host mimicry we built a sequence-based scan sliding 7-mers of NS1 against six human proteins implicated in cross-reactivity, scored against a shuffled-window null. It could not recover mimicry relationships already documented in the literature, so we discarded its output rather than report a negative result from an instrument that fails its own positive control.

6. The proposed test

6.1 The format

The literature review establishes why a quantitative deployable NS1 test is missing. This section specifies what to build, in enough detail to argue with.

The proposal is a homogeneous, mix-and-read bioluminescent assay. A fingerstick sample of no more than 50 μL is added to a single tube holding lyophilised reagents: both switch arms and the substrate furimazine. NS1 binds both arms at once, bringing LgBiT and SmBiT into proximity and reconstituting an active luciferase. A smartphone camera in a light-tight adapter integrates the photons over a fixed exposure, and the app converts photon count to a concentration. As said before, this requires one tube, one addition, no washes, no plate reader.

The physical components are unremarkable, which is sort of the point. Lyophilised protein reagents are the standard route to ambient-stable diagnostics, though

furimazine's stability in that form is unresolved. What matters about the adapter is fixed geometry rather than optical quality, since quantification depends on the sample-to-sensor distance being identical between calibration and measurement, and the app must use fixed exposure and ISO.

6.2 Why the format follows from the architecture

The homogeneous format is really not a stylistic preference, and rather it follows directly from putting *recognition* and *reporting* in the same molecule.

Every laboratory requirement of an ELISA exists to separate bound antibodies from the unbound. In the process, the capture antibody is immobilised so unbound material can be washed away, and the detection antibody exists because the capture antibody cannot report. Now, a proximity switch needs none of this, because unbound arms emit no light. Discrimination happens within the molecule rather than on a surface.

Table 7. Requirement-by-requirement comparison of a quantitative NS1 ELISA with the proposed format.

ELISA, the current quantitative standard	Proposed format
Capture antibody immobilised on a plate	No solid phase
Wash steps to remove unbound sample	No washes
Separate labelled detection antibody	Reporter built into the binder
Plate reader	Phone camera
Cold chain for antibody reagents	Lyophilised, ambient
Trained operator, laboratory setting	One addition, at the point of care

We chose bioluminescence over fluorescence deliberately, because fluorescence needs an excitation source and an emission filter, and blood autofluoresces. Bioluminescence, on the other hand, makes its own photons, so the reader collapses to a bare camera sensor in a dark enclosure.

6.3 Quantification

Raw photon count is not a concentration. It varies with temperature, since luciferase activity is strongly temperature-dependent, and with substrate age, reaction time, sample volume, phone model and haematocrit. A single-channel reading from an uncontrolled consumer device will not give a reliable number, and this is where most bioluminescent point-of-care proposals become vague.

Three approaches increase in robustness. A fixed calibration curve stored in the app is simplest and most fragile. An internal spike control corrects for reagent and reaction variation but not optical geometry. The recommended option is a ratiometric dual-colour readout: a spectrally distinct reference luciferase, reporting the ratio of two emission channels rather than absolute intensity. The ratio is insensitive to exposure, distance,

reagent quantity and temperature, because both channels vary together, and published camera-read sensors quantify this [19, 24] way.

The cost is real: a second reporter must be accommodated in the construct design, and the two emissions must separate on an unmodified phone's RGB channels. The reference luciferase then doubles as a reaction control, resolving a failure a quantitative test cannot tolerate: a dark tube is invalid, not negative. A sample-adequacy check completes the set.

6.4 What would kill it, and what it would cost

Four failure modes are serious enough to state plainly.

The hook effect is the most serious. At high antigen concentration each arm binds a separate antigen molecule instead of both binding the same one, cross-linking collapses, and signal falls as concentration rises, so a high sample reads low. NS1 spans roughly three orders of magnitude in circulation, so this is not hypothetical. Running each sample at two dilutions and flagging disagreement is the standard mitigation, but the effect must be characterised before any clinical claim.

Blood absorbs blue light. NanoLuc emits near 460 nm [22], where haemoglobin absorbs strongly, and the attenuation varies with haematocrit, exactly the patient-to-patient variation that destroys quantification. A plasma-separation membrane removes the problem at the cost of a step; a red-shifted variant would too, but compatibility with split complementation at the required affinity is unresolved.

NS1 circulates as a hexamer, and this cuts both ways. Multivalency should raise apparent affinity, but the geometry in Section 5 was measured on the monomer, a clean one-to-one regime. On a hexamer, additional bridging modes exist that were never modelled: a genuine gap between the structural work and the assay proposal.

Background complementation sets the fourth limit. Raising arm concentration to improve sensitivity also raises the intrinsic-affinity floor, so an optimum exists that only a titration can locate.

A quantitative NS1 measurement today requires an ELISA: a plate, a reader, cold-chain reagents, and a trained operator, meaning a district hospital laboratory and a sample-transport step. The proposed format replaces that capital equipment with a phone the user already owns and a moulded adapter, moving the recurring cost into the consumable.

The format requires no component that does not already exist. What it requires is that the switch constructs work, which is what has not been tested.

7. Discussion

7.1 Interpretation

The three questions this project set out to answer came back differently: one qualified yes, one clear yes, and one no that turned out to matter more than either.

The first has to be split, because its halves came out differently. On conservation, the footprint we can actually evidence is the wing / domain-I site, not the β -ladder the question named. That one is 84% conserved across DENV1-4, verified against the alignment rather than assumed from serotype labels. For the β -ladder we checked serotype conservation across all 45 positions of 261-305, but deposited no Zika table, so the flavivirus half of the question is unanswered there. At the site we did characterise, discrimination is available but not automatic, because four positions differ between dengue and Zika. On the design itself the answer is qualified. Candidates engage the intended region reliably across seeds, but we cannot claim a converged pose: different seeds place the same binder on overlapping, non-identical footprints.

Second, two binders can occupy non-overlapping sites on NS1 with geometry compatible with split-reporter complementation. This is the claim least dependent on any predictor: it rests on distances measured between coordinates, which is why it survives the third answer intact.

For the third, standard co-folding confidence metrics do not identify correct binders on this target. That is stronger than a null result: the metric is anti-correlated with the property it is supposed to track, so a shortlist ranked on it selects against epitope fidelity.

Three explanations can account for the inversion, and we can rank them without resolving between them. The first is target-specific miscalibration: these metrics were benchmarked largely on globular protein-protein interfaces, and a flat β -sheet face may fall outside the regime where they carry the signal. The second is intrinsic target difficulty: a rigid extended sheet offers little of the shape complementarity these models associate with binding. The third is a sampling limit. The evidence favours the first two: the decoy that never touches the epitope scores highest of anything tested.

7.2 Limitations

The limitations majorly fall into three classes:

Scope (decided before execution): Narrowing the biomarker panel from six candidates to NS1 alone brought depth at the cost of any cross-target claim: nothing here establishes that the calibration result generalises beyond this antigen. The wing-domain direction was carried as a stretch objective and never run as a full campaign. And every prediction used the NS1 monomer, though the antigen circulates as a hexamer, so the epitope presented physiologically may differ from the one designed against.

Technical (met during execution): Five seeds is thin for estimating a rate, so per-design frequencies should be read as ordinal rather than precise. The replacement ranking criterion was defined after seeing the calibration data, so it describes the binders

we observed rather than independently testing them. And every pose here comes from a predictor we showed will place a published antibody on the wrong face of this target, so reproducibility establishes consistency, not correctness.

Deployment (arising only afterwards): Host mimicry, the reason the tip region is excluded, is not resolvable computationally: sequence-level scans failed their own positive control here, and structural avoidance is a proxy for an immunological question only an immunoassay answers. More fundamentally, no computational method can establish that a proximity switch switches. Signal magnitude, dynamic range and background complementation require the constructs to exist, so the geometry here is a necessary condition for complementation rather than evidence of it.

Corrections made during the project: Technical claims and literature-scoping claims alike were retracted on measurement rather than left standing- the full ledger, with the evidence that triggered each change, is in Appendix A12.

7.3 Future work

The next steps are sequenced, and the first is the project's promised deliverable.

1. **Build the switch constructs (weeks 1 to 2, no new compute):** The specification in Section 5.3 fixes the construct space, and writing each combination out is a text operation on measured geometry.
2. **Add the control set (same period):** Free unfused arms, a non-complementing same-epitope pair, reporter-only background, and a linker-length series. Without these a null result cannot be distinguished from a construction error.
3. **Re-run the controls on a second predictor (about four GPU-hours, in parallel):** AF2-multimer should face the same 2B7 and 3G2 controls that Boltz-2 failed. Either it recovers the published epitopes, giving a trustworthy ternary model, or it does not, establishing that no available tool provides one. The second outcome is the more useful.
4. **One wet-lab screen (partner-dependent):** Binding on the free arms by BLI or SPR, complementation on the assembled constructs by luminescence, with dose-response across the clinical range.
5. **A dedicated second-arm campaign (after step 4 reports):** Target the wing and domain-I site while deliberately engaging the four positions that differ between dengue and Zika.

Priority follows cost: steps 1 to 3 need no partner and no new experimental resource, which is why they come first. The *provisional target product profile* in the Appendix sets the numbers any resulting device would have to meet. It is author-proposed rather than established, and its central row, a predicted advantage in secondary infection, is stated with the three reasons it might not hold.

7.4 Confidence and what it would take to change our mind

Our confidence differs sharply between the three findings.

The calibration result is the most secure. It rests on 15 predictions with a crystallographic positive control and a non-cognate decoy, both behaving as an answer key rather than a benchmark we tuned against, and it reproduces at design level across 10 designs. What would overturn it is a second predictor recovering the published epitopes where this one did not, step 3 above, localising the failure to one tool rather than the metric class.

The geometry is secure differently, because superposition and clash detection do not depend on a confidence score. What would overturn it is the structures themselves being wrong, which the wet-lab screen tests directly.

The binder candidates are the least secure claim here as they are characterised, not validated, and a binding assay could refute them in a week.

8. Conclusion

We set out to design a protein switch that binds dengue NS1 and reports how much of it is there, turning a yes-or-no field test into a number. That remains the goal, and this report covers the half of it we could complete with confidence.

What exists now: two characterised binder arms at non-overlapping sites on NS1, one at the conserved β -ladder patch and one at the wing and domain-I face, whose contact footprint is 84% conserved across DENV1-4. A measured guarantee that they occupy those sites simultaneously without colliding, from distances between coordinates rather than any model's confidence, which therefore survives the failure below. A construction-ready specification fixing linker lengths and reporter placement. And a scoring harness with the calibration that justifies it, released openly.

What does not exist: any experimental validation, and the assembled switch. The second is the more immediate case. We had planned to model the switch before building it, and the calibration control showed the predictor placing a solved antibody on the wrong face of this target in ten runs out of ten. A model that cannot locate a published epitope cannot adjudicate a designed one, so we stopped ranking on it rather than produce a number we would not believe.

That is why the next step is a wet-lab bench and not another GPU. Express the two arms, measure binding to recombinant NS1, and the cheapest experiment in the programme either falsifies this work or licenses the rest of it.

In conclusion, the field test tells you whether, but we are building the one that tells you how much, and what stands between the two is now specified.

Author Contributions

In general contributions:

- Kaeyura did the computational part of the project and formulated the impact chain

- Rabaani did the literature review, and the proposed diagnostic test
- Kaeyura and Rabaani did the pivots and decisions throughout

1. Introduction - Kaeyura

2. Literature review - Rabaani

3. Theory of change - Rabaani, Kaeyura created the figure

4. Methodology - Kaeyura

5. Results - Kaeyura

6. The proposed test - Rabaani

7. Discussion - Kaeyura

8. Conclusion - Rabaani

References - Equal Contribution

AI Disclosure

We used Claude (Anthropic), the Claude Science agent platform (Anthropic), and Gemini (Google) during this project. Every use is listed below.

- **Running the compute pipeline:** We installed the Baker Lab design tools (RFdiffusion, ProteinMPNN, and related structure-prediction tools) on a shared drive on RunPod GPU hosts. Because individual runs took hours and often failed for trivial reasons, we used Claude Science to orchestrate them: restarting failed runs, launching the next run once one finished, and shutting the GPU machines down at the end. We decided what to run and why, and the agent managed execution.
- **Code and analysis debugging:** We used Claude science to troubleshoot our scoring harness and structure-analysis scripts (contact analysis, epitope-fidelity scoring, conservation checks), explain errors, and check our reasoning about metrics such as ipTM and interface pAE.
- **Figure generation:** We used Claude to write and debug the matplotlib code that turns our results tables into the report figures. The data, analysis, and figure design were originally ours or were generated by our experiments.
- **Polishing:** After we finished writing, we used Claude and Gemini to polish the text by tightening wording and fixing grammar. We wrote every passage ourselves. The models edited existing text and did not generate any.

We designed the study, ran the analysis, and reached the conclusions ourselves. The report's structure is our own. No original analysis, arguments, findings, or conclusions were AI-generated.

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Appendix

This appendix contains the complete per-seed data behind every result in Section 5, the DENV1-4 alignment and target product profile that support the surrounding claims, and a corrections ledger of everything withdrawn on measurement. A13 gives the harness settings, contact definitions, and join keys required to reproduce any reported figure from the deposited files.

A1 Design shortlist

Table A1. Design shortlist: all ten candidates under both rankings, with the rank shift for each.

label	pipeline_id	target_site	on_epitope_seeds_of_5	mean_epitope_contacts	mean_2B7_overlap_of_20	mean_ipT_M	rank_by_epitope_fidelity	rank_by_mean_ipT_M	rank_shift
B1	onepi_pd_5_dldesign_1	β-ladder	5	6.2	9.6	0.379	1	9	8
B2	onepi_pd_16_dldesign_0	β-ladder	4	4.8	7.6	0.344	2	10	8
B3	onepi_pd_5_dldesign_0	β-ladder	2	2.6	3.6	0.405	3	8	5
B4	onepi_pd_15_dldesign_0	β-ladder	1	2.2	2.8	0.578	4	5	1
B5	onepi_pd_20_dldesign_1	β-ladder	1	1.6	2.2	0.591	5	4	-1
B6	onepi_pd_7_dldesign_1	β-ladder	1	1.4	2.2	0.478	6	7	1
W1	seed5_pd_11_dldesign_0	wing / domain-I	1	1.4	2	0.713	7	1	-6

label	pipeline_id	target_site	on_epitope_seeds_of_5	mean_epitope_contacts	mean_2B7_overlap_of_20	mean_ipT_M	rank_by_epitope_fidelity	rank_by_mean_ipT_M	rank_shift
W2	seed5_pipeline_id_18_design_1	wing / domain-I	1	1.4	2.4	0.492	8	6	-2
B7	onepi_pipeline_id_30_design_1	β -ladder	1	1.2	1.6	0.648	9	3	-6
W3	seed5_pipeline_id_15_design_1	wing / domain-I	0	0	0	0.675	10	2	-8

Source: design_shortlist_table.csv. Labels follow §4.2: B = β -ladder site, W = wing / domain-I site, numbered by epitope-fidelity rank within each site.

pipeline_id is the join key for every deposited CSV and appears here only so the deposit can be navigated. rank_shift is (rank by ipTM) minus (rank by epitope fidelity). Positive means the design is better than its confidence score suggests. The two rankings share no molecules in their top three.

The on_epitope_seeds_of_5 and mean_epitope_contacts columns are measured against the β -ladder window (261-305) and report the epitope-window rate of Section 4.2, not the stricter primary criterion of Section 4.3. For the three wing / domain-I designs this window is not their target, so W3's zero is the expected reading for a molecule binding elsewhere, not a failure. W3's own footprint is characterised in A7.

A2 Gate calibration:

Table A2.1. Gate calibration: all fifteen predictions, per seed, with measured contact sets.

construct	role	seed	ipT_m	interface_pAE_A	complex_pldd_t	gate_passes	beta_ladder_contacts_MEASURED	clusterA_hotspots	binding_site_MEASURED	all_contacts_MEASURED	overlap_2B7_of_20
nb_3 (design)	design	1	0.643	7.03	0.878	fail	7	269	beta-ladder (ON-EPITOPE)	28 92 93 269 270 281 299 303 304 305 324 325 326 327 328 346	12
nb_3 (design)	design	2	0.515	9.15	0.878	fail	6		beta-ladder (ON-EPITOPE)	28 93 94 95 270 281 301 303 304 305 307 326 327 328	10
nb_3 (design)	design	3	0.778	3.43	0.91	PASS	0		N-terminus decoy	1 2 3 4 6 8 10 11 12 13 15 28 29 190 214 324	0
nb_3 (design)	design	4	0.713	5.56	0.86	fail	0		wing/connector (off-epitope)	35 36 37 38 39 40 74 110 113 115 116 118 121 122 160 162 163 168	0
nb_3 (design)	design	5	0.752	5.29	0.893	fail	6	269	beta-ladder (ON-EPITOPE)	269 281 299 303 304 305 306 324 325 326 327 328 346	11

construct	role	seed	ip tm	interface_pAE_A	complex_plddt	gate_pas s	beta_ladder_contacts_MEASURED	clusterA_hotspots	binding_site_MEASURED	all_contacts_MEASURED	overlap_2B7_of_20
2B7 (crystal ref)	validated protective Ab	1	0.478	6.15	0.943	fail	10	269	beta-ladder (ON-EPITOPE)	269 270 281 282 299 301 302 303 304 305 307 325 326 327 328 330 343 344 345 346	20
2B7 (crystal ref)	validated protective Ab	2	0.461	6.48	0.944	fail	10	269	beta-ladder (ON-EPITOPE)	269 270 281 282 299 301 302 303 304 305 307 325 326 327 328 330 343 345 346	19
2B7 (crystal ref)	validated protective Ab	3	0.444	6.56	0.943	fail	10	269	beta-ladder (ON-EPITOPE)	269 270 281 282 299 301 302 303 304 305 307 326 327 328 330 343 345 346	18
2B7 (crystal ref)	validated protective Ab	4	0.462	6.48	0.942	fail	10	269	beta-ladder (ON-EPITOPE)	269 270 281 282 299 301 302 303 304 305 307 326 327 328 330 343 344 345 346	19
2B7 (crystal ref)	validated protective Ab	5	0.455	6.48	0.941	fail	10	269	beta-ladder (ON-EPITOPE)	269 270 281 282 299 301 302 303 304 305 307 326 327 328 330 343 344 345 346	19
cAb-Lys3 (NEGATIVE)	non-cognate NEGATIVE	1	0.881	2.23	0.884	PASS	0		N-terminus / dimer interface	1 2 3 4 5 6 8 14 184 185 186 187 188 189 228 229 230 231	0
cAb-Lys3 (NEGATIVE)	non-cognate NEGATIVE	2	0.83	3.44	0.852	PASS	0		N-terminus / dimer interface	1 2 3 4 5 6 7 8 9 16 17 18 19 20 184 185 186 187 188 189 228 229 230 231	0
cAb-Lys3 (NEGATIVE)	non-cognate NEGATIVE	3	0.766	5.05	0.832	fail	0		N-terminus / dimer interface	1 2 4 5 6 7 8 9 14 17 18 19 184 185 186 187 188 189 228 229 230 231	0
cAb-Lys3 (NEGATIVE)	non-cognate NEGATIVE	4	0.823	3.58	0.863	PASS	0		N-terminus / dimer interface	1 2 4 5 6 7 8 14 16 17 18 20 185 186 187 188 189 228 229 230 231	0
cAb-Lys3 (NEGATIVE)	non-cognate	5	0.673	6.36	0.858	fail	0		N-terminus / dimer interface	1 2 4 5 6 7 8 14 18 20 184 185 186 187 188	0

construct	role	seed	ip tm	interface_pAE_A	complex_plddt	gate_passes	beta_ladder_contacts_MEASURED	clusterA_hotspots	binding_site_MEASURED	all_contacts_MEASURED	overlap_2B7_of_20
ACTIVE)	NEGATIVE									189 190 228 229 230 231	

Source: gate_calibration_multiseed.csv. Three constructs, five seeds each, one harness. beta_ladder_contacts_MEASURED counts heavy-atom contacts within 4.5 Å falling inside 261-305. gate_pass records the three confidence components of the Section 4.3 gate (pLDDT, ipTM, interface pAE) and not the contact requirement, which is why a construct can be marked passing with zero epitope contacts; both columns are shown so the discrepancy is visible rather than resolved silently. overlap_2B7_of_20 counts how many of the twenty residues in the 2B7 crystallographic contact set each prediction recovers, so the reference recovers 18 to 20 of its own 20 across seeds while the non-cognate decoy recovers none.

Per-construct summary:

Table A2.2. Gate calibration summarised per construct

Construct	Role	Mean ipTM	SD	Gate passes	Mean β-ladder contacts
2B7	Protective antibody, crystal reference (PDB 6WER)	0.460	0.012	0 / 5	10.0
nb_3	Our design	0.680	0.105	1 / 5	3.8
cAb-Lys3	Non-cognate decoy (PDB 1MEL)	0.795	0.079	3 / 5	0.0

mean ipTM, standard deviation, gate passes and mean β-ladder contacts across five seeds. Spearman $\rho(\text{ipTM}, \beta\text{-ladder contacts}) = -0.888$, $p = 9.7 \times 10^{-6}$, $n = 15$. Gate-passing predictions average 0.0 epitope contacts; gate-failing predictions average 6.27.

A3 Predictor mislocation:

Table A3. Predictor mislocation

run	construct	role	seed	ip tm	interface_pae	plddt	nb3_epitope_contacts	nb3_contacts	fab_beta_ladder_hits	fab_own_epitope_overlap_of_16	verdict
ctrl_3g2_s1	ctrl_3g2	arm control (G2 + NS1)	1	0.483	6.57	0.906	n/a (nb_3 not in construct)		6	0	ARM CONTROL FAIL: 3G2 recovers 0/16 of its own cryo EM epit

run	con str uct	ro le	s e e d	i p t m	interf ace_p ae	pl d dt	nb3_epit ope_cont acts	nb3_contacts	fab_beta ladder_ hits	fab_own_epit ope_overlap_ of_16	ver dict
											ope; land s on beta -lad der
ctrl_ 3g2_ s2	ctrl_ _3g 2	ar m co nt rol (3 G2 + N S1)	2	0. 6 3 4	4.56	0. 91 4	n/a (nb_3 not in construct)		6	0	AR M CO NT RO L FAI L: 3G2 reco vers o/1 6 of its own cryo EM epit ope; land s on beta -lad der
ctrl_ 3g2_ s3	ctrl_ _3g 2	ar m co nt rol (3 G2 + N S1)	3	0. 41 3	7.21	0. 9 0 4	n/a (nb_3 not in construct)		5	0	AR M CO NT RO L FAI L: 3G2 reco vers o/1 6 of its own cryo EM epit ope; land s on beta -lad der
ctrl_ 3g2_ s4	ctrl_ _3g 2	ar m co nt rol (3 G2	4	0. 5 8 6	5.36	0. 91	n/a (nb_3 not in construct)		5	0	AR M CO NT RO L FAI

run	construct	role	seed	iptm	interface	pldt	nb3_epitope_contacts	nb3_contacts	fab_beta_ladder_hits	fab_own_epitope_overlap_of_16	verdict
		+ N S1)									L: 3G2 reco vers o/1 6 of its own cryo EM epit ope; land s on beta -lad der
ctrl_3g2_s5	ctrl_3g2	arm control (3 G2 + N S1)	5	0.45	7	0.918	n/a (nb_3 not in construct)		6	0	ARM CONTROL FAI L: 3G2 reco vers o/1 6 of its own cryo EM epit ope; land s on beta -lad der
tern_full_s1	tern_full	ternary (nb_3 + 3G2 + N S1)	1	0.488	5.73	0.913	0	2;3;4;6;8;11;13;188;206;210;232;234;235;253;254	6	0	INV ALI D: 3G2 occ upie s the beta -lad der (nb _3's epit ope) , disp laci ng nb_

run	con str uct	ro le	s e e d	i p t m	interf ace_p ae	pl d dt	nb3_epit ope_cont acts	nb3_contacts	fab_beta ladder_ hits	fab_own_epit ope_overlap_ of_16	ver dict
											3 to a dec oy -- co-b indi ng can not be asse ssed
tern _full _s2	tern _ful l	ter na ry (n b_ 3 + 3G 2 + N S1)	2	0. 3 7 5	9.08	0. 8 8 4	0	33;35;36;37;39;70;71;74;113;115; 116;118;119;120;157;162;163;164; 166;168	5	0	INV ALI D: 3G2 occ upie s the beta -lad der (nb _3's epit ope) , disp laci ng nb_ 3 to a dec oy -- co-b indi ng can not be asse ssed
tern _full _s3	tern _ful l	ter na ry (n b_ 3 + 3G 2 + N S1)	3	0. 4 0 2	7.42	0. 8 97	0	2;3;4;6;10;11;12;13;15;188;189;21 0	6	0	INV ALI D: 3G2 occ upie s the beta -lad der (nb _3's epit ope)

run	con str uct	ro le	s e e d	i p t m	interf ace_p ae	pl d dt	nb3_epit ope_cont acts	nb3_contacts	fab_beta ladder_ hits	fab_own_epit ope_overlap_ of_16	ver dict
											, displacing nb_3 to a decoy -- co-binding cannot be assessed
tern_full_s4	tern_full	ternary (nb_3 + 3G2 + N S1)	4	0.367	9.11	0.893	0	33;35;36;37;38;39;71;74;115;116;118;119;120;122;123;156;157;162;163;168	5	0	INV ALI D: 3G2 occupies the beta-ladder (nb_3's epitope), displacing nb_3 to a decoy -- co-binding cannot be assessed
tern_full_s5	tern_full	ternary (nb_3 + 3G2 +	5	0.435	8.33	0.895	0	35;36;37;38;39;71;74;113;115;116;118;119;120;122;123;124;160;161;162;168	5	0	INV ALI D: 3G2 occupies the beta-lad

run	construct	role	seed	ip_tm	interface_pae	plddt	nb3_epitope_contacts	nb3_contacts	fab_beta_ladder_hits	fab_own_epitope_overlap_of_16	verdict
		NS1)									der (nb_3's epitope), displacing nb_3 to a decoy -- co-binding cannot be assessed

Source: ternary_results.csv. 3G2 predicted against NS1, ten runs. fab_own_epitope_overlap_of_16 counts how many of the antibody's sixteen cryo-EM epitope residues the prediction recovered.

Recovery was 0 of 16 in 10 of 10 runs. Across those ten runs ipTM ranged 0.37 to 0.63 and pLDDT 0.88 to 0.92.

Two alternative explanations were tested rather than argued. Only 1 of the 16 epitope residues lies within 5 Å of a disordered or absent region, so the epitope was available to be found. It is not buried in the NS1 dimer interface, so it was not occluded in the form predicted.

A4 Steric re-feasibility against intact NS1

Table A4. Steric re-feasibility against intact NS1: all forty-two designs, with clashing residues, atom pairs below 2.0 Å, and buried surface area per side.

design	campaign	n_hotspots_cropped	hotspots	epitope_contacts_cropped	overlap_2B7	clashing_NS1_residues_fulllength	atom_pairs_under_2Å	sterically_feasible	buried_per_side_A2
onepi_pd_18_dldesign_o	RF2_cropped_target	3	269;294;296	18	8	10	47	False	659
onepi_pd_19_dldesign_o	RF2_cropped_target	3	269;294;296	15	5	15	92	False	799
onepi_pd_20_dldesign_o	RF2_cropped_target	3	269;294;296	17	6	11	117	False	832
onepi_pd_28_dldesign_o	RF2_cropped_target	3	269;294;296	22	6	12	123	False	907

design	campaign	n_hotspots_cropped	hotspots	epitope_contacts_cropped	overlap_B7	clashing_NS1_residues_fulllength	atom_pairs_under_2A	sterically_feasible	buried_per_side_A2
seed5_pd_18_dldesign_1	RF2_cropped_target	3	269;274;296	26	11	17	134	False	1367
seed5_pd_19_dldesign_0	RF2_cropped_target	3	269;272;296	17	8	16	137	False	719
seed5_pd_12_dldesign_1	RF2_cropped_target	3	269;294;296	13	3	27	145	False	670
onepi_pd_12_dldesign_0	RF2_cropped_target	3	269;294;296	22	7	12	150	False	796
seed5_pd_19_dldesign_1	RF2_cropped_target	4	269;272;294;296	23	7	20	153	False	986
seed5_pd_16_dldesign_0	RF2_cropped_target	3	269;294;296	20	10	18	156	False	1048
seed5_pd_5_dldesign_0	RF2_cropped_target	2	269;296	14	6	23	165	False	753
onepi_pd_16_dldesign_0	RF2_cropped_target	5	269;272;274;294;296	26	9	22	180	False	1012
seed5_pd_10_dldesign_0	RF2_cropped_target	2	294;296	26	5	22	188	False	1460
seed5_pd_23_dldesign_1	RF2_cropped_target	2	269;296	15	6	27	203	False	710
onepi_pd_12_dldesign_1	RF2_cropped_target	2	294;296	25	8	17	207	False	1187
onepi_pd_5_dldesign_0	RF2_cropped_target	4	269;272;294;296	27	9	30	209	False	1138
onepi_pd_7_dldesign_1	RF2_cropped_target	4	269;272;294;296	29	11	19	212	False	1222
onepi_pd_20_dldesign_1	RF2_cropped_target	5	269;272;274;294;296	33	11	23	232	False	1689
seed5_pd_2_dldesign_0	RF2_cropped_target	2	269;296	16	8	25	234	False	816
seed5_pd_11_dldesign_0	RF2_cropped_target	3	269;272;296	31	11	21	239	False	1497
seed5_pd_2_dldesign_1	RF2_cropped_target	2	269;296	16	8	28	244	False	794
onepi_pd_30_dldesign_1	RF2_cropped_target	4	269;274;294;296	34	11	30	256	False	1438

design	campaign	n_hotspots_cropped	hotspots	epitope_contacts_cropped	overlap_2B7	clashing_NS1_residues_fulllength	atom_pairs_under_2A	sterically_feasible	buried_percent_side_A2
seed5_pd_1_dldesign_0	RF2_cropped_target	3	269;294;296	16	6	33	277	False	643
onepi_pd_24_dldesign_1	RF2_cropped_target	4	269;272;294;296	18	8	33	281	False	893
onepi_pd_11_dldesign_0	RF2_cropped_target	3	269;272;296	24	8	40	350	False	933
onepi_pd_5_dldesign_1	RF2_cropped_target	3	269;274;296	34	11	30	355	False	1900
seed5_pd_3_dldesign_0	RF2_cropped_target	3	269;294;296	19	5	48	396	False	816
seed5_pd_15_dldesign_1	RF2_cropped_target	4	269;272;274;296	23	8	50	406	False	1054
seed5_pd_20_dldesign_1	RF2_cropped_target	4	269;272;294;296	19	2	50	416	False	675
onepi_pd_15_dldesign_0	RF2_cropped_target	5	269;272;274;294;296	38	11	44	462	False	1703
onepi_pd_9_dldesign_0	RF2_cropped_target	3	269;272;296	24	8	59	566	False	999
seed5_pd_4_dldesign_1	RF2_cropped_target	4	269;272;294;296	19	2	79	874	False	748
P2_21_d2	P2_fulllength	0		8	1	n/a (built on full-length)		yes	n/a
P2_21_d1	P2_fulllength	0		8	1	n/a (built on full-length)		yes	n/a
P2_0_d1	P2_fulllength	0		7	1	n/a (built on full-length)		yes	n/a
P2_0_d2	P2_fulllength	0		7	1	n/a (built on full-length)		yes	n/a
P2_5_d2	P2_fulllength	0		5	1	n/a (built on full-length)		yes	n/a
P2_5_d1	P2_fulllength	0		5	1	n/a (built on full-length)		yes	n/a
P2_97_d1	P2_fulllength	0		4	2	n/a (built on full-length)		yes	n/a
P2_89_d1	P2_fulllength	0		4	1	n/a (built on full-length)		yes	n/a
P2_89_d2	P2_fulllength	0		4	1	n/a (built on full-length)		yes	n/a
P2_73_d1	P2_fulllength	0		3	1	n/a (built on full-length)		yes	n/a

Source: shortlist_refeasibility.csv. Designs generated against a cropped target, then re-scored against full-length NS1. Burial is computed here with FreeSASA on the deposited complexes and reported as the mean of the target-side and binder-side values, which is the convention under which the 2B7 crystal reference gives 602 Å²; Appendix A8 uses target-side burial throughout instead, where the same reference gives 597 Å².

Of the 32 designs from the cropped-target campaign, zero were sterically feasible on the intact target, with 47 to 874 atom pairs closer than 2.0 Å. The design ranked first in that campaign buried 1,703 Å² per side against 602 Å² for the 2B7 crystallographic reference, which is the signature of a binder occupying volume that the intact protein already fills. The buried_per_side_A2 column reports this for all 32, computed with FreeSASA as the mean of target-side and binder-side burial; the largest value in the campaign is 1,900 Å². Note that the burial convention matters when comparing against A8: the 2B7 reference is 602 Å² per side but 597 Å² target-side, and A8 reports target-side throughout.

A5 Rescue campaign: all fifty predictions

Table A5. Rescue campaign: all fifty predictions against full-length NS1, ten designs at five seeds each.

design	campaign	seed	idm	epitope_contacts_261_305	hotspots_engaged	overlap_2B7_of_20	total_contacts	on_epitope	all_contacts
onepi_pd_15_dlde_sign_o	onepi	1	0.27	0		0	11	False	85;87;89;124;125;126;127;128;129;130;132
onepi_pd_15_dlde_sign_o	onepi	2	0.8	0		0	17	False	2;3;4;5;6;8;11;12;13;28;29;160;161;162;163;164;165
onepi_pd_15_dlde_sign_o	onepi	3	0.612	0		0	21	False	35;36;37;38;39;71;74;113;115;116;118;119;120;123;159;160;161;162;163;166;168
onepi_pd_15_dlde_sign_o	onepi	4	0.662	11	269;272;296	14	31	True	62;66;69;92;93;94;95;138;139;140;141;269;270;272;296;297;298;299;302;303;304;305;326;327;330;336;342;343;344;345;346
onepi_pd_15_dlde_sign_o	onepi	5	0.543	0		0	20	False	35;36;37;39;71;74;113;115;116;118;119;120;122;123;161;162;163;164;166;168
onepi_pd_16_dlde_sign_o	onepi	1	0.204	5	269	9	11	True	92;93;269;270;299;303;304;326;327;330;346
onepi_pd_16_dlde_sign_o	onepi	2	0.516	7	272	9	14	True	92;94;138;272;299;301;302;303;304;305;306;326;327;328
onepi_pd_16_dlde_sign_o	onepi	3	0.139	6	269	10	13	True	92;137;138;269;299;301;302;303;305;326;327;330;346
onepi_pd_16_dlde_sign_o	onepi	4	0.552	0		0	19	False	4;5;6;8;11;13;21;22;23;26;27;28;29;31;162;192;201;203;214
onepi_pd_16_dlde_sign_o	onepi	5	0.307	6	269	10	13	True	94;138;139;269;270;299;301;303;304;327;330;345;346

design	campaign	seed	ipmt	epitope_contacts_261_305	hotspots_engaged	overlap_2B7_of_20	total_contacts	on_epitope	all_contacts
onepi_pd_20_dlde sign_1	onepi	1	0.476	0		0	17	False	37;38;39;40;74;113;114;115;116;118;120;122;157;160;162;166;168
onepi_pd_20_dlde sign_1	onepi	2	0.773	0		0	22	False	35;36;37;38;39;41;110;113;114;115;118;119;120;121;157;160;161;162;163;164;166;168
onepi_pd_20_dlde sign_1	onepi	3	0.511	0		0	21	False	33;35;36;37;38;39;40;70;71;74;110;113;114;115;118;119;157;160;164;166;168
onepi_pd_20_dlde sign_1	onepi	4	0.446	8	272	11	18	True	92;94;137;138;270;272;281;299;302;303;304;305;324;326;327;328;329;346
onepi_pd_20_dlde sign_1	onepi	5	0.751	0		0	21	False	35;36;37;39;71;74;113;114;115;116;118;119;120;122;157;160;161;162;163;166;168
onepi_pd_30_dlde sign_1	onepi	1	0.576	0		0	20	False	35;37;38;39;74;113;114;115;117;118;119;157;158;159;160;161;162;163;166;168
onepi_pd_30_dlde sign_1	onepi	2	0.767	0		0	25	False	35;36;37;38;39;40;41;74;110;113;114;115;116;117;118;119;122;157;159;160;161;162;163;166;168
onepi_pd_30_dlde sign_1	onepi	3	0.701	0		0	17	False	2;3;4;5;6;8;11;12;13;15;20;22;26;27;28;29;192
onepi_pd_30_dlde sign_1	onepi	4	0.664	0		0	20	False	2;3;4;6;13;15;16;17;18;19;20;22;158;160;161;162;163;164;165;189
onepi_pd_30_dlde sign_1	onepi	5	0.534	6	274	8	19	True	26;27;28;29;30;62;274;281;300;301;304;305;306;307;324;326;327;328;329
onepi_pd_5_dlde sign_o	onepi	1	0.586	0		0	17	False	33;35;73;74;77;115;118;119;120;121;123;124;161;162;163;164;166
onepi_pd_5_dlde sign_o	onepi	2	0.614	0		0	22	False	33;34;35;70;71;73;74;110;113;114;115;118;120;124;157;158;159;160;161;162;166;168
onepi_pd_5_dlde sign_o	onepi	3	0.488	0		0	28	False	33;34;35;37;38;39;40;41;67;70;71;73;74;111;112;113;114;115;116;117;118;120;122;156;158;159;166;168
onepi_pd_5_dlde sign_o	onepi	4	0.132	7	272	9	12	True	28;272;281;299;301;303;304;305;306;326;327;328
onepi_pd_5_dlde sign_o	onepi	5	0.205	6		9	10	True	281;299;301;303;304;305;306;326;327;328

design	campaign	seed	iptm	epitope_contacts_261_305	hotspots_engaged	overlap_2B7_of_20	total_contacts	on_epitope	all_contacts
onepi_pd_5_dlidesign_1	onepi	1	0.383	6	269;272	7	18	True	62;92;93;94;95;138;218;219;269;270;271;272;299;303;325;326;327;346
onepi_pd_5_dlidesign_1	onepi	2	0.463	7	269	12	16	True	92;138;269;270;299;302;303;304;305;306;327;340;343;344;345;346
onepi_pd_5_dlidesign_1	onepi	3	0.165	5	269	9	16	True	69;92;95;126;138;140;269;270;299;303;304;325;326;327;328;346
onepi_pd_5_dlidesign_1	onepi	4	0.567	8		12	13	True	28;281;282;299;301;302;303;304;305;307;326;327;328
onepi_pd_5_dlidesign_1	onepi	5	0.319	5		8	9	True	299;302;303;304;305;306;326;327;346
onepi_pd_7_dlidesign_1	onepi	1	0.584	0		0	13	False	89;90;121;122;123;124;125;126;127;128;129;130;132
onepi_pd_7_dlidesign_1	onepi	2	0.396	0		0	25	False	33;35;36;37;38;39;70;71;74;110;112;113;115;118;119;120;121;123;157;158;160;161;162;166;168
onepi_pd_7_dlidesign_1	onepi	3	0.539	0		0	22	False	33;35;36;37;38;39;40;41;70;71;74;110;111;113;115;118;119;157;160;161;162;168
onepi_pd_7_dlidesign_1	onepi	4	0.528	7	269	11	18	True	92;94;138;139;140;269;270;299;301;303;304;305;306;326;327;328;329;346
onepi_pd_7_dlidesign_1	onepi	5	0.341	0		0	29	False	33;35;36;37;38;39;40;70;71;74;77;78;110;111;112;113;114;115;116;117;118;119;120;124;157;160;162;166;168
seed5_pd_11_dlidesign_0	seed5	1	0.8	0		0	22	False	33;35;69;70;72;73;74;76;88;89;90;91;92;113;115;118;119;121;122;123;125;126
seed5_pd_11_dlidesign_0	seed5	2	0.756	0		0	25	False	33;35;69;70;72;73;74;76;89;90;91;92;113;115;116;118;119;120;122;123;124;125;126;128;163
seed5_pd_11_dlidesign_0	seed5	3	0.389	0		0	37	False	3;4;5;6;8;10;11;13;15;20;22;23;24;26;27;28;29;30;31;32;158;160;161;162;163;165;183;184;185;189;190;192;196;197;198;201;203
seed5_pd_11_dlidesign_0	seed5	4	0.886	0		0	27	False	33;35;66;69;70;71;72;73;74;76;87;89;90;91;113;115;116;118;119;120;123;124;125;126;128;139;163
seed5_pd_11_dlidesign_0	seed5	5	0.734	7		10	12	True	281;299;300;301;303;304;305;326;327;328;329;346

design	campaign	seed	ipm	epitope_contacts_261_305	hotspots_engaged	overlap_2B7_of_20	total_contacts	on_epitope	all_contacts
seed5_pd_15_dlde_sign_1	seed5	1	0.671	0		0	22	False	35;36;37;38;39;74;113;115;116;118;119;120;122;123;157;158;160;161;162;163;166;168
seed5_pd_15_dlde_sign_1	seed5	2	0.713	0		0	23	False	35;36;37;38;39;74;78;113;115;116;118;119;120;122;123;157;158;160;161;162;163;166;168
seed5_pd_15_dlde_sign_1	seed5	3	0.703	0		0	23	False	33;35;36;37;38;39;74;113;114;115;116;118;119;120;122;123;157;160;161;162;163;166;168
seed5_pd_15_dlde_sign_1	seed5	4	0.741	0		0	22	False	33;35;37;38;39;40;74;113;115;116;118;119;120;122;123;157;160;161;162;163;166;168
seed5_pd_15_dlde_sign_1	seed5	5	0.545	0		0	26	False	8;9;11;33;35;37;38;39;70;74;113;114;115;116;118;119;122;123;157;160;161;162;163;164;166;168
seed5_pd_18_dlde_sign_1	seed5	1	0.469	0		0	19	False	36;37;38;39;74;113;115;116;118;119;120;123;157;160;161;162;163;166;168
seed5_pd_18_dlde_sign_1	seed5	2	0.483	0		0	21	False	36;37;38;39;74;113;115;116;118;119;120;156;157;158;159;160;161;162;163;166;168
seed5_pd_18_dlde_sign_1	seed5	3	0.541	0		0	15	False	36;37;113;115;116;117;118;119;120;122;123;124;160;162;168
seed5_pd_18_dlde_sign_1	seed5	4	0.557	0		0	20	False	35;36;37;38;39;74;113;115;116;118;119;123;157;159;160;161;162;163;166;168
seed5_pd_18_dlde_sign_1	seed5	5	0.41	7		12	14	True	28;281;282;299;302;303;304;305;306;307;326;327;330;346

Source: rescue_results.csv. The on_epitope column is the epitope-window test of Section 4.2 (≥ 4 contacts in 261-305), not the primary criterion of Section 4.3, which additionally requires two hotspots including one from cluster A. Two of these fifty predictions meet the primary criterion: B1 at seed 1 and B4 at seed 4. These sequences were generated against a cropped target and re-co-folded here freely against full-length NS1, so the poses differ from those tested in A4.

A6 Charge-variant series

Table A6. Charge-variant series: four variants across four CDR net-charge states, five seeds each, testing whether charge drives epitope engagement.

variant	arm	CDR_net_charge	epitope_contacts	hotspots	overlap_2B7	on_epitope
v_q1_s1	v_q1	1	0		0	False
v_q1_s2	v_q1	1	8	269	12	True
v_q1_s3	v_q1	1	8	269	14	True
v_q1_s4	v_q1	1	0		0	False

variant	arm	CDR_net_charge	epitope_contacts	hotspots	overlap_2B7	on_epitope
v_q1_s5	v_q1	1	7		13	True
v_q2_s1	v_q2	2	0		0	False
v_q2_s2	v_q2	2	8	269;274	11	True
v_q2_s3	v_q2	2	0		0	False
v_q2_s4	v_q2	2	0		0	False
v_q2_s5	v_q2	2	7		11	True
v_q3_s1	v_q3	3	0		0	False
v_q3_s2	v_q3	3	7	269;296	10	True
v_q3_s3	v_q3	3	0		0	False
v_q3_s4	v_q3	3	4	269	9	True
v_q3_s5	v_q3	3	8	269	15	True
v_q5_s1	v_q5	5	3	272;274	5	False
v_q5_s2	v_q5	5	3	269	7	False
v_q5_s3	v_q5	5	0		0	False
v_q5_s4	v_q5	5	4	269	8	True
v_q5_s5	v_q5	5	3		6	False

Source: charge_variant_results.csv. Four variants across four CDR net-charge states, five seeds each for twenty predictions, testing whether charge drives epitope engagement.

Spearman ρ (CDR net charge, epitope contacts) = -0.137 , $p = 0.564$, $n = 20$. Per charge state: +1 max 8 contacts (mean 4.6), +2 max 8 (mean 3.0), +3 max 8 (mean 3.8), +5 max 4 (mean 2.6). No relationship. The hypothesis and the recommendation built on it are withdrawn; see A12.

A7 Second-arm conservation against DENV1-4

Table A7. Second-arm conservation: the nineteen-residue contact consensus of design W3, position by position, across DENV1-4 and Zika.

position	in_planned_EP2	DEN V1	DEN V2	DEN V3	DEN V4	ZI KV	conserved_D ENV1-4	zika_divergent	discriminating	dimer_occluded
35	no	Q	Q	Q	Q	H	YES	YES	YES	no
37	no	D	E	D	E	D	no	no	no	no
38	no	S	S	S	S	S	YES	no	no	no
39	no	P	P	P	P	P	YES	no	no	no
74	yes	E	E	E	E	E	YES	no	no	no
113	yes	Y	Y	Y	Y	H	YES	YES	YES	no
115	yes	W	W	W	W	W	YES	no	no	no
116	no	K	K	K	K	K	YES	no	no	no
118	yes	W	W	W	W	W	YES	no	no	no
119	yes	G	G	G	G	G	YES	no	no	no
122	yes	K	K	K	K	Y	YES	YES	YES	no
123	yes	I	M	I	I	F	no	no	no	no
157	no	D	D	D	D	D	YES	no	no	no
160	no	F	F	F	F	F	YES	no	no	yes
161	no	G	G	G	G	G	YES	no	no	no
162	no	I	V	V	M	V	no	no	no	no
163	no	F	F	F	F	F	YES	no	no	no
166	no	N	N	N	N	S	YES	YES	YES	no
168	no	W	W	W	W	W	YES	no	no	no

Source: second_arm_conservation.csv. The nineteen-residue contact consensus of design W3, position by position, across all four dengue serotypes and Zika.

Sixteen of nineteen positions (84%) are conserved across DENV1-4. Four positions differ between dengue and Zika, so discrimination against the nearest cross-reactive flavivirus is available but requires deliberately engaging those positions, which the current design does not.

Two conservation figures appear in this report and they are not the same measurement. The 84% above is the **realised** footprint, measured from the predicted complex. The 75% quoted at design time (6 of 8) was the **planned** epitope window, chosen before any structure existed. The in_planned_EP2 column shows which positions belong to each.

A8 Direction B burial analysis

Table A8. Direction B burial analysis: fifteen designed poses plus the 2B7 crystallographic reference and one cropped-target control, target-side burial throughout.

design	s e e d	epitope_buri al_A2_261_3 05	tip_burial _A2_311_ 352	total_target_ side_burial_ A2	epitope_fra ction_of_to tal	passes_ep itope_ge3 oo	passes_ _tip_le 40	passe s_bot h	burial_c onventi on
onepi_pd_1 5_dl design_0	4	413	341	1140	0.36	True	False	False	target-side, BL-filtered
onepi_pd_30_ dl design_1	5	228	201	857	0.27	False	False	False	target-side, BL-filtered
onepi_pd_20_ dl design_1	4	321	292	818	0.39	True	False	False	target-side, BL-filtered
onepi_pd_5_ dl design_1	1	226	229	791	0.29	False	False	False	target-side, BL-filtered
onepi_pd_16_ dl design_0	2	312	231	706	0.44	True	False	False	target-side, BL-filtered
onepi_pd_7_ dl design_1	4	316	134	675	0.47	True	False	False	target-side, BL-filtered
onepi_pd_5_ dl design_1	3	194	142	662	0.29	False	False	False	target-side, BL-filtered
onepi_pd_5_ dl design_1	2	309	159	631	0.49	True	False	False	target-side, BL-filtered
onepi_pd_16_ dl design_0	5	316	162	596	0.53	True	False	False	target-side, BL-filtered

design	seed	epitope_burial_A2_261_305	tip_burial_A2_311_352	total_target_side_burial_A2	epitope_fraction_of_total	passes_epitope_geo300	passes_tip_le40	passes_both	burial_convention
onepi_pd_5_dldesign_1	4	246	176	515	0.48	False	False	False	target-side, BL-filtered
onepi_pd_16_dldesign_0	1	235	155	513	0.46	False	False	False	target-side, BL-filtered
onepi_pd_5_dldesign_1	5	238	119	449	0.53	False	False	False	target-side, BL-filtered
onepi_pd_5_dldesign_0	4	226	149	405	0.56	False	False	False	target-side, BL-filtered
onepi_pd_16_dldesign_0	3	119	91	338	0.35	False	False	False	target-side, BL-filtered
onepi_pd_5_dldesign_0	5	190	151	355	0.54	False	False	False	target-side, BL-filtered
2B7 (crystal 6WER, protective Ab)	-	320	250	597	0.54	True	False	False	target-side, BL-filtered
nb_3 (seed 5, CROPPED target 255-310)	5	356	n/a - tip absent from construct	375	0.95	True	untestable	untestable	target-side, BL-filtered; total is whole 255-310 fragment, not comparable to full-length rows

Source: direction_b_burial.csv. Fifteen designed poses plus the 2B7 crystallographic reference and one cropped-target control. Burial here is target-side, so the 2B7 reference reads 597 Å² where Table A4 reports 602 Å² per side for the same structure. The tip window is defined as 311-352 by sequence, but the structure used (9W22 chain A) is ordered only to residue 346, so every tip burial reported here is measured over 311-346.

The pre-registered safety criterion required ≥ 300 Å² of epitope burial (261-305) and ≤ 40 Å² of tip burial (311-352), the tip being avoided for host-mimicry reasons. Six of fifteen designs met the epitope threshold; **none** met the tip threshold, minimum observed 91 Å². The two burials are positively correlated across the fifteen designs (Pearson $r = 0.706$, $p = 0.0033$), because epitope and tip are interdigitated strands of one β -sheet.

The decisive test is the reference: 2B7 buries 320 Å² of epitope and 250 Å² of tip, failing the criterion while being a real, protective antibody. The criterion is therefore refuted rather than unmet, and is withdrawn.

A9 Switch geometry

Table A9. Switch geometry: all twenty-five seed-pairs, with inter-binder distance, clash count and terminus separations.

arm1_seed	arm2_seed	NS1_CA_RM SD_A	min_interbinder _dist_A	clash_pairs _lt2A	centroid_s ep_A	N1_N 2_A	N1_C 2_A	C1_N 2_A	C1_C 2_A
s1	s1	3.65	2.78	0					
s1	s2	3.68	5.32	0					
s1	s3	2.85	0.53	4					
s1	s4	3.6	1.82	1					
s1	s5	3.58	0.55	25					
s2	s1	1.33	14.24	0	46.7	77	61	49.1	40.5
s2	s2	1.31	15.9	0	48.3	77.9	64.7	49.8	42.6
s2	s3	1.65	8.99	0	44.3	75.6	55.2	47.1	36.6
s2	s4	1.09	12.65	0	46	77.4	59.5	48.7	39.2
s2	s5	3.95	4.05	0					
s3	s1	3.94	9.85	0					
s3	s2	3.97	12.29	0					
s3	s3	3.66	4.85	0					
s3	s4	3.9	9.01	0					
s3	s5	1.94	2.42	0	35.6	63.9	44.9	47.5	28.5
s4	s1	1.63	8.38	0	43.1	65.1	52.6	57.8	26.7
s4	s2	1.68	9.81	0	45	66.3	56.4	59.3	30.6
s4	s3	2	7.54	0	40.9	64.7	46.8	56.8	21.9
s4	s4	1.32	8.72	0	42.8	66.3	51.3	58	25.5
s4	s5	4.05	5.64	0					
s5	s1	4.24	15.33	0					
s5	s2	4.22	17.13	0					
s5	s3	3.55	12.92	0					
s5	s4	4.11	15.37	0					
s5	s5	2.17	10.96	0	49.1	42.9	16.1	75.7	51.2

Source: switch_geometry_measurements.csv. Every combination of five arm-1 seeds and five arm-2 seeds, modelled on the same NS1 target.

Twenty-two of twenty-five pairs clear the 2.0 Å steric threshold, and all ten well-converged pairs do. Well-converged means the two predictions' NS1 copies superpose within 2.2 Å Cα RMSD: the ten qualifying pairs range 1.09 to 2.17 Å, the fifteen excluded 2.85 to 4.24 Å. These are not superpositions of a fixed target onto itself. Each prediction folds NS1 independently alongside its binder, so target coordinates differ run to run, and two copies disagreeing by 4 Å cannot support a terminus measurement between them. C-terminus separations are reported only for the ten converged pairs: median 33.6 Å over all ten, or 30.6 Å over the nine excluding a 51.2 Å outlier, range 21.9-42.6 Å. The closest converged pair sits at 2.42 Å minimum inter-binder distance with zero atom pairs under 2.0 Å: it clears the threshold, but by the smallest margin of any converged pair.

These are measurements on predicted complexes, not predictions of switch behaviour.

A10 DENV1-4 alignment

Source: ns1_aln.fasta. Four sequences, 352 ungapped columns, offset 0. DENV2 (the design target) is 335 of 346 identical to the consensus, 96.8%.

Every conservation claim in this report is checked against this alignment rather than assumed from serotype labels. The alignment file is deposited in full; A7 gives the per-position readout for the residues that matter.

A11 Provisional target product profile

Analytical performance

Table A11.1. Provisional target product profile, analytical performance.

Parameter	Minimal	Optimal	Basis
Analyte	DENV NS1 antigen, native conformation	same	Design constraint: a de novo binder recognises a 3D conformational surface, so any heat/acid dissociation step is closed to us [P]
Dynamic range	1-1,000 ng/mL	0.1-10,000 ng/mL	NS1 circulates roughly ng/mL-μg/mL [L]; commercial-grade sensors span 1-200 and 0.01-1,000 ng/mL [L]
Limit of detection	≤ 1 ng/mL	≤ 0.1 ng/mL	Existing platforms reach 22 pg/mL (0.022 ng/mL) in undiluted human serum (DNA aptamer electrochemical, serotype 4) [L]; a low-cost FTO aptasensor reaches 0.21 ng/mL in PBS and 0.55 ng/mL in serum [L]. Femtogram-per-millilitre figures also appear in this literature but are largely buffer-phase, so the serum-validated 22 pg/mL is the bar we set against, and the targets here sit deliberately above it in exchange for equipment-free operation
Quantitative precision	CV ≤ 25% across the range	CV ≤ 15%	[P]. Must resolve a day-to-day change on serial sampling; a 25% CV distinguishes a roughly 2-fold change, which is the clinically meaningful unit for kinetics
Clinical sensitivity	≥ 87% vs RT-PCR	≥ 95%	Beat the best fielded NS1 RDT (Abbott, 87.4%) [L];

Parameter	Minimal	Optimal	Basis
			the fielded range is 61.0-87.4% [L]
Clinical specificity	≥ 97%	≥ 99%	Do not regress: fielded NS1 RDTs achieve 97-100% [L]
Serotype coverage	Detects DENV1-4	equivalent sensitivity across all four	Verified per-position against the DENV1-4 alignment, never assumed from sequence [P]
Flavivirus specificity	No cross-reaction with ZIKV NS1 at clinically relevant concentrations	quantitatively discriminates ZIKV	A dual dengue/Zika aptasensor could not distinguish the two analytes [L]. A demonstrated failure mode, not a theoretical worry
Cross-reactivity, other febrile illness	Negative in enteric fever, malaria, Japanese encephalitis, leptospirosis	same	Fielded NS1 assays already achieve this [L]
Performance in secondary infection	Sensitivity gap vs primary infection < 10 points	no measurable gap	[P]. The project's central mechanistic bet (see below)

Proposed specification for the test described in §6. Every row is marked [L] literature, [P] proposed by us, or [U] unknown, so a reader can separate evidence from intention.

Operational performance

Table A11.2. Provisional target product profile, operational performance.

Parameter	Minimal	Optimal	Basis
Sample type	Fingerstick whole blood or plasma	whole blood, no separation	[P]; LUMABS-class sensors work directly in blood plasma [L]
Sample volume	≤ 50 µL	≤ 10 µL	[P]. Fingerstick-compatible
Assay format	Homogeneous, mix and read	same	Architectural consequence: the split reporter needs no capture surface, no wash steps, no secondary antibody
Time to result	≤ 30 min	≤ 10 min	[P]. Single-visit result
Equipment	Consumer smartphone camera, no potentiostat, no plate reader, no fabricated electrodes	smartphone only	[P]; camera readout of bioluminescence is established for antibodies and small molecules, not yet for a viral antigen in whole blood [U]
Power	Battery or none	none	[P]
Operator training	≤ 1 day	none beyond pictorial instructions	[P]
Reagent storage	Ambient to 30 °C, no cold chain	ambient to 40 °C	[P]
Shelf life	≥ 12 months at ambient	≥ 24 months	[P]

Parameter	Minimal	Optimal	Basis
Cost per test	Comparable to an NS1 RDT	below	[P]. A cost the product must meet; not yet estimated. A BOTEK belongs here before this row is asserted [U]

Marking convention as in Table A11.1

Two honest gaps in this table. The cost row carries no bottom-up estimate, so it should be read as an aspiration rather than a costing. And the claim that no published target product profile exists for quantitative NS1 monitoring rests on a preliminary search; it needs verifying against WHO, FIND and PAHO before it is asserted anywhere else.

A12 Corrections ledger

Claims made earlier in this project and subsequently withdrawn on measurement. They are listed because a benchmark that hides its own corrections is not a benchmark.

Table A12. *Corrections ledger: claims made earlier in this project and withdrawn on measurement, with what replaced each and the evidence that forced the change.*

#	Claim as originally made	What replaced it	Evidence
1	All five β -ladder hotspots cannot be engaged by one paratope	Refuted; single paratopes do engage the hotspot set	reranking_finding.md
2	A burial ceiling near $370 \text{ \AA}^2$ limits epitope engagement	Withdrawn; observed epitope burial reaches $413 \text{ \AA}^2$	A8
3	CDR net charge is the single remaining structural lever for epitope engagement	Withdrawn; $\rho = -0.137$, $p = 0.564$, $n = 20$	A6
4	The $\geq 300 \text{ \AA}^2$ epitope / $\leq 40 \text{ \AA}^2$ tip criterion is achievable but unmet	Refuted, not unmet; the crystallographic reference itself fails it	A8
5	3G2 / 4B8 as the primary second arm	Retracted on conservation (31% and 31% across DENV1-4); retained as methods-development models only	sandwich_partner_finding.md
6	Second-arm conservation reported as 0 of 19	Corrected to 16 of 19 after a residue-numbering offset was found	A7
7	Dimeric NS1 presentation improves on-epitope rate (registered prediction)	Refuted; monomer 3/5 on-epitope against dimer 1/5	Job 970e8822-b1d0-4465-9377-0436adbof065
8	A near-contact seed-pair showed 14 atom pairs under $3.0 \text{ \AA}$	Unsupported; the table records zero pairs under $2.0 \text{ \AA}$ and has no $3.0 \text{ \AA}$ column	A9

#	Claim as originally made	What replaced it	Evidence
9	Binder success rates in the literature average 19% at 10 μ M across five targets	Retracted; the figure was not supported by any source consulted	

Items 1, 2 and 4 share a cause: each was a structural impossibility claim inferred from a small sample rather than measured. Item 6 is the one that would have been most damaging if it had survived, because it would have invalidated the second arm.

A13 Reproduction instructions

Deposited files. All eleven data tables above, the DENV1-4 alignment, the target structure (target_9W22_chainA.pdb), the scoring harness, and the finding documents that record each result as it was obtained.

Joining the tables: Every CSV is keyed by pipeline_id (for example onepi_pd_5_dldesign_1), not by the B/W labels used in the report body. A1 is the mapping table between the two. Any analysis that starts from a report label must pass through A1 first.

The harness: One configuration for every prediction reported here: Boltz-2 2.2.1, recycling_steps 3, diffusion_samples 1, --use_msa_server, --no_kernels, seeds 1 through 5. No prediction in this report was produced under different settings, and none was rerun after inspection.

Contact definition: A residue-residue contact is a heavy-atom pair within 4.5 Å. Every binding site, epitope-contact count, hotspot engagement and clash assignment is computed at this cutoff from coordinates, never assumed from design intent: a design targeted at the β -ladder is credited with epitope contacts only if the predicted structure puts them there. A separate 2.0 Å cutoff is used only for the steric-clash test.

Numbering: All residue positions refer to DENV2 NS1 as deposited in PDB 9W22 chain A, residues 1-346, with alignment offset 0. The offset was verified against the DENV1-4 alignment after an earlier numbering error produced a conservation figure that was wrong by sixteen positions; see ledger item 6.

A14 Cross-validator agreement

Table A14. Cross-validator agreement: five designs scored under both predictors with identical inputs.

	AF2_ipTM	AF2_pLDDT	AF2_iPAE_mean_A	Boltz_ipTM	Boltz_pLDDT	Boltz_iPDE_A	hotspots_8A	hot
nn1	0.16	0.586	22.3	0.484	0.737	6.04	2	[27]
nn2	0.20	0.554	21.9	0.497	0.731	7.18	2	[26]
nn	0.14	0.57	24.0	0.481	0.7	6.19	2	[27]
	0.18	0.701	22.5	0.454	0.803	6.65	0	[]
	0.16	0.67	23.0	0.226	0.827	8.52	1	[27]

Source: multi_validator_results.csv.

Five designs from an earlier campaign, each re-scored under AlphaFold2-multimer and Boltz-2 with identical inputs, to test whether the calibration result of Section 5.1 was one model's idiosyncrasy. These are not the ten designs ranked in A1: they are three β -ladder backbones (BC_l73_mpnn1, BC_l73_mpnn2, BC_l107_mpnn1), the nb_3 construct used as the design arm of the gate calibration, and a PD-1 binder carried as a control. AF2 returns ipTM 0.14 to 0.20 across all five; Boltz-2 returns 0.23 to 0.50 on the same structures. None clears the ipTM > 0.7 gate under either predictor, and the two rank the same molecules differently, so the inversion of Section 5.1 is not an artefact of a single model. Hotspot columns are Boltz-2 poses at an 8 Å cutoff and are reported for orientation only.

A15 Monomer versus dimer: a registered prediction

Table A15. Monomer versus dimer target presentation: five seeds each, one harness.

target_form	seed	iptm	interface_pAE_A	complex_plddt	beta_ladder_contacts	binding_site_MEASURED	cross_protomer
MONOMER (control)	1	0.643	7.03	0.878	7	beta-ladder (ON-EPITOPE)	
MONOMER (control)	2	0.515	9.15	0.878	6	beta-ladder (ON-EPITOPE)	
MONOMER (control)	3	0.778	3.43	0.91	0	N-terminus decoy	
MONOMER (control)	4	0.713	5.56	0.86	0	wing/connector (off-epitope)	
MONOMER (control)	5	0.752	5.29	0.893	6	beta-ladder (ON-EPITOPE)	
DIMER	1	0.739	5.63	0.882	0	wing/connector (off-epitope)	YES
DIMER	2	0.731	5.58	0.892	0	wing/connector (off-epitope)	no
DIMER	3	0.723	5.52	0.892	0	N-terminus decoy	YES
DIMER	4	0.724	5.69	0.888	0	N-terminus decoy	YES
DIMER	5	0.769	4.64	0.894	9	beta-ladder (ON-EPITOPE)	no

Source: dimer_revalidation.csv.

Before running it we registered the prediction that presenting NS1 as the physiological dimer rather than the monomer would improve on-epitope frequency, on the reasoning that the dimer presents the β -ladder as the continuous sheet it forms in solution. It did not. The monomer control is on-epitope in 3 of 5 seeds and the dimer in 1 of 5, so the prediction is refuted and the monomer is retained throughout this report. Note also that dimer ipTM is uniformly higher (0.72 to 0.77 against 0.52 to 0.78 for the monomer) while epitope contact is lower, which is the same inversion seen in the gate calibration.